

ORE Studio User Manual

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Part I

Getting Started

Everything you need to go from a fresh installation to a fully provisioned, operational OreStudio deployment: what the application is, how to connect to a backend, and how to bootstrap the system for first use.

Introduction

About ORE Studio

ORE Studio is a desktop application for exploring, configuring, and running quantitative finance calculations. It provides a graphical environment built around the *Open Source Risk Engine (ORE)*, a widely-used open-source library for pricing derivatives and measuring financial risk, which is itself built on [QuantLib](#). ORE Studio handles the data — storing trades, market data, and model configurations, and presenting the results — while ORE and QuantLib provide the mathematics.

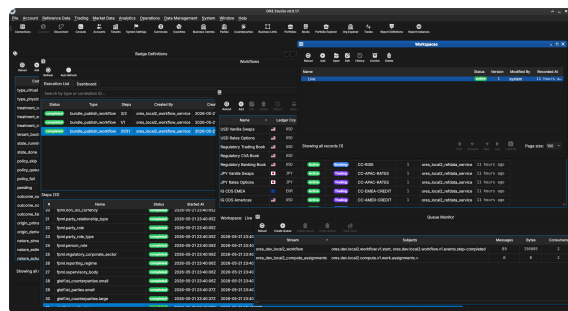


Figure 1.1: ORE Studio v0.0.17 — the main workspace showing the instrument and trade management views.

Audience

This manual is written for several kinds of reader:

- **Analysts, traders, quants, and middle-office staff** who want to explore financial instruments and risk calculations through a graphical interface. No programming experience is required.
- **Students and researchers** learning quantitative finance. ORE Studio makes it possible to experiment with derivative pricing, yield curve construction, and risk measures without writing code.

- **LLM-based agents and coding assistants** that interact with ORE Studio on a user's behalf, help configure workspaces, interpret results, or draft analytical reports. Multimodal models — those capable of processing both text and images — are preferred, as the manual makes extensive use of screenshots to describe the user interface.

The focus throughout is on what you can do through the application itself: entering trades, configuring market data, running analytics, and understanding the results.

Functional Areas

ORE Studio is organised around a set of functional areas accessible from the main menu:

- *Reference data* — currencies, business calendars, counterparties, and market conventions that form the foundation for everything else.
- *Instruments and trades* — define, store, and manage financial instruments and their terms.
- *Market data* — yield curves, volatility surfaces, fixing histories, and other inputs to pricing and risk models.
- *Model configuration* — choose and parameterise pricing engines, simulation models, and sensitivity specifications.
- *Analytics* — run calculations and view results: net present value, sensitivities (Greeks), credit and funding valuation adjustments (XVA), Monte Carlo simulations, and stress tests.

All data is stored persistently, so trades, curves, and configurations can be saved, revisited, and reused across sessions.

What ORE Studio Is Not

ORE Studio is a learning and exploration environment, not a production trading or risk system. It is independent of and unaffiliated with ORE, QuantLib, or any financial institution. All quantitative mathematics are provided by ORE and QuantLib; ORE Studio is the surface through which they are configured and their results explored. Users who need production-grade performance, real-time market data feeds, or regulatory reporting should look to enterprise risk platforms.

How This Manual Is Organised

The chapters follow the natural order of first use. After this introduction, *Connecting to ORE Studio* covers launching the application, establishing a connection to the backend, and orienting yourself in

the main window; *Initial Setup* then covers provisioning the system, tenants, and parties. Later chapters cover each functional area in turn. You do not need to read the manual cover to cover; each chapter can be read independently once the system is up and running.

Early-Stage Software Notice

Early-stage software. ORE Studio is currently in active development (version 0.x). Features, workflows, and this documentation are all evolving and may change between releases. Numbers produced by the system are for learning and exploration only — they must not be used for real trading, risk management, or any financial decision-making.

A Note on This Document

This manual was generated entirely by large language models (LLMs) working under human supervision. While every effort has been made to ensure accuracy, LLM-generated content can contain errors, omissions, or descriptions that do not match the actual software behaviour. If you find a discrepancy between this manual and the application, trust the application. Please report inaccuracies via the project issue tracker so they can be corrected.

Connecting to ORE Studio

Getting started

When you launch ORE Studio a splash screen appears briefly while the application initialises, showing the ORE Studio logo and the build version in its lower-right corner.

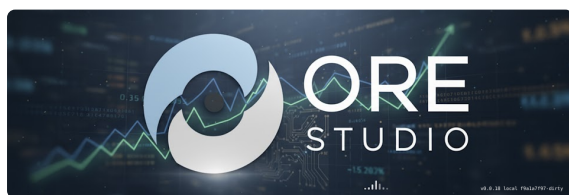


Figure 2.1: The ORE Studio splash screen shown during start-up. The build version appears in the lower-right corner.

Once initialisation completes you are greeted by the main window in its *disconnected* state. The application is running but has not yet established a connection to the backend service that performs calculations. Before you can work with trades, market data, or analytics, you need to connect to a running ORE Studio backend.

This chapter covers:

- What the main window looks like before a connection is established.
- How to open the Connection Manager and configure one or more backends.
- How to log in once a connection is active.
- How to manage your saved connections — adding, editing, deleting, organising by environment and tag.
- How to read the status bar and understand what each zone shows.
- How to launch ORE Studio from the command line with options for telling windows apart, configuration directory, auto-connect, and diagnostics.

The main window before login

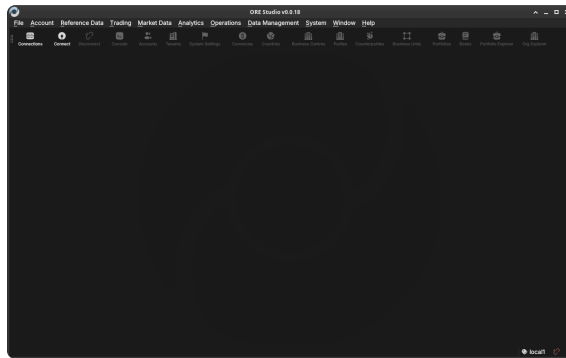


Figure 2.2: The ORE Studio main window immediately after launch, in the disconnected state. The toolbar shows the Connect and Connections buttons; the workspace area is empty until you log in.

In the disconnected state most of the application is inactive. The menu bar and toolbar are present but workspace panels, trade lists, and analytics views are all unavailable until a successful login. Two toolbar buttons are always accessible:

- **Connect** — opens the login dialog for the currently selected connection.
- **Connections** — opens the Connection Manager, where you configure the list of backends ORE Studio knows about.

The Connection Manager

The Connection Manager is the central place where you maintain the list of ORE Studio backends the application can connect to. You might have a local development backend, a shared team backend, and a staging backend — each appears as a separate entry here. The manager lets you add, edit, delete, and organise those entries before you attempt a login.

Open it at any time from the toolbar **Connections** button or from the *File* menu.

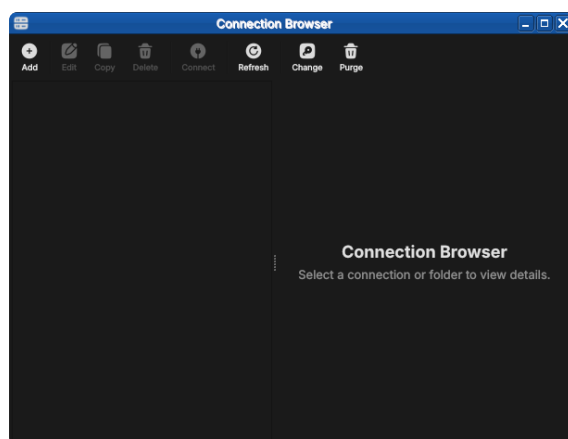


Figure 2.3: The Connection Browser open with no connections configured. The toolbar exposes Add, Edit, Copy, Delete, Connect, Refresh, Change and Purge; the unusable actions are greyed out until something is selected.

Adding a new connection

Click **Add** (or the "+" button) to open the *New Connection* form. Fill in the fields:

- **Name** — a free-text label you choose, used throughout the UI to identify this backend (e.g. "Local dev", "Team staging"). Must be unique among your saved connections.
- **Host** — the hostname or IP address of the machine running the ORE Studio backend (e.g. localhost, 192.168.1.10, ores-staging.example.com).
- **Port** — the TCP port the backend is listening on. The default is 35900; change it only if the backend was started on a different port.
- **Environment** — an optional grouping label (see [Environments](#) below). Helps you distinguish development, staging, and production backends at a glance.
- **Tags** — zero or more free-text labels (see [Tags](#) below). Useful for filtering and searching when you have many connections.

Figure 2.4: The New Connection form with example values filled in. *Type* is set to *Connection*; the *Environment* may be left as *manual entry* or linked to a defined environment.

Click **Save** (or **OK**) to add the connection to the list. The new entry appears immediately in the Connection Manager list. No network contact is made at this point — ORE Studio simply stores the configuration.

Editing a connection

Select an existing connection in the list and click **Edit** (or double-click the row) to open the same form pre-populated with the saved

values. Change any field and click **Save**. The connection list updates immediately.

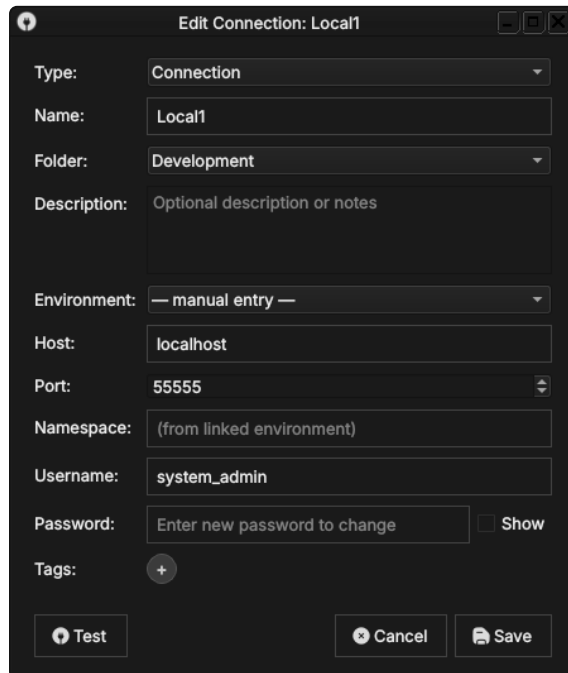


Figure 2.5: Editing a saved connection. The dialog title shows *Edit Connection*, the fields are pre-populated, and the password field reads *Enter new password to change* — the stored password is preserved unless you type a replacement.

Deleting a connection

Select one or more connections in the list and click **Delete** (or press the Delete key). A confirmation dialog asks you to confirm the removal.

Deleting a connection removes only the saved configuration; it has no effect on the running backend or any data stored there.

Organising connections into folders

When the list grows, *folders* keep it manageable. A folder is a named container you can nest, grouping related connections — by team, by client, or by environment tier. In the populated browser above, the connections sit under a *Development* → *Dev* → *Local1...Local4* folder tree.

To create one, click **Add** and set *Type* to *Folder*. Give it a name and an optional description, and choose a parent folder (or *No Folder* to place it at the top level). Connections are then assigned to a folder through their *Folder* field, or by dragging them in the tree.

Protecting your credentials with a master password

The passwords you save with a connection are encrypted at rest using a *master password* that you choose. The first time you save a credential, ORE Studio offers to create one.

Type the same value into *New Password* and *Confirm Password*; the fields turn green when they match. *Show password* reveals what you

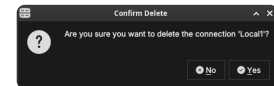


Figure 2.6: The delete confirmation dialog. ORE Studio names the connection being removed and waits for you to confirm with *Yes* or cancel with *No*.

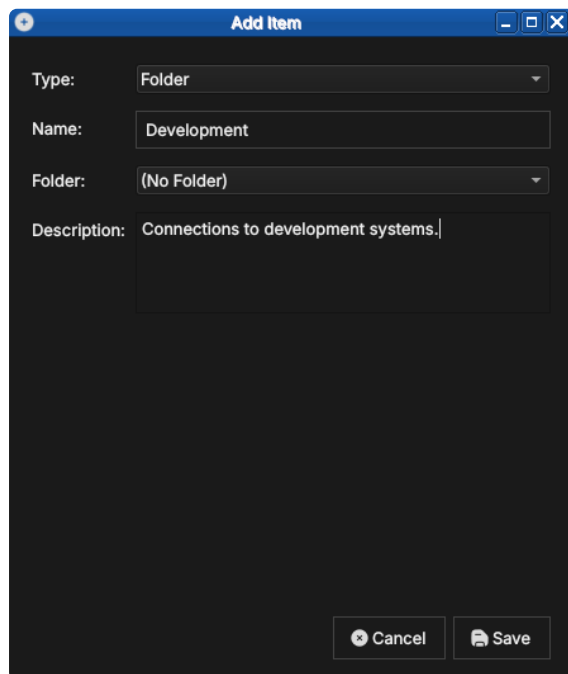


Figure 2.7: Creating a folder. With *Type* set to Folder, a folder needs only a name and an optional description; it can be nested inside another folder.

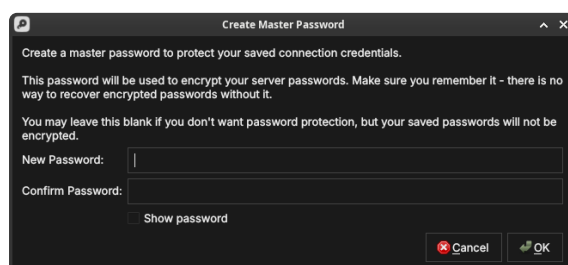


Figure 2.8: Creating the master password. It encrypts every saved server password; there is no recovery if you forget it, so you may also leave it blank to store passwords unencrypted.

typed.

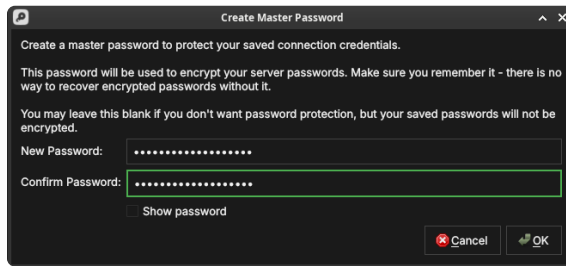


Figure 2.9: The master password confirmed — both fields match and are outlined in green, ready to accept with OK.

When you next launch ORE Studio and open the Connection Browser, it prompts you to unlock your saved connections by entering the master password. Until you do, the encrypted passwords stay sealed.

If you leave the master password blank, your connections still work but their passwords are stored unencrypted — acceptable on a personal machine, but not recommended on shared or portable systems.

Environments

An *environment* is a named grouping that you assign to a connection to indicate which tier of your infrastructure it belongs to. Typical values are Development, Staging, and Production, but the list is fully customisable — you can define any environment names that make sense for your setup.

Environments appear as a coloured badge or label next to the connection name in the list, making it immediately obvious which tier you are about to connect to. This is a safety feature: it is easy to accidentally connect to production when you meant staging; a clearly labelled environment badge prevents that.

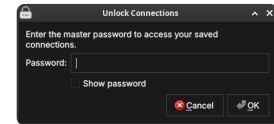


Figure 2.10: The unlock prompt shown on a later launch. Enter the master password to decrypt your saved connection credentials.

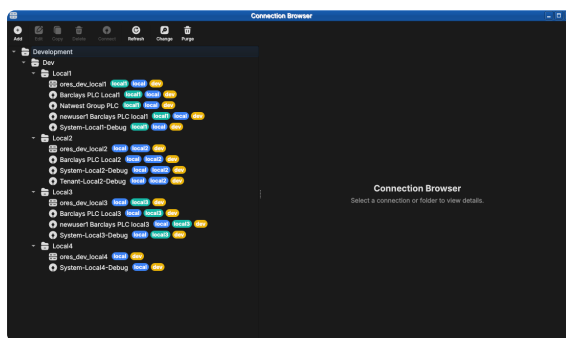


Figure 2.11: A populated Connection Browser. Connections are grouped under folders (here Development → Dev → Local1...Local4) and each carries coloured chips — the environment and tags — making the tier and purpose of every entry obvious at a glance.

To manage the available environment names, open the *Environments* section within the Connection Manager settings (or the dedicated menu item). From there you can:

- **Add** a new environment name and choose its display colour.
- **Rename** an existing environment (all connections using it update automatically).

- **Delete** an environment (connections that used it revert to *Unsigned*).

An environment is itself created from the Connection Browser: click **Add** and set *Type* to *Environment*. An environment carries its own host, port, HTTP port, namespace and tags — these become the connection details that any connection linked to it inherits.

The screenshot shows the 'Add Item' dialog box with the following fields and values:

- Type:** Environment (dropdown)
- Name:** ores.dev.local1
- Folder:** Development (dropdown)
- Description:** Local1 development environment
- Host:** localhost
- Port:** 52001
- HTTP Port:** 8080
- Namespace:** ores.dev.local1
- Tags:** dev (chip with close button) and a plus button to add more tags
- Buttons:** Cancel and Save at the bottom right

Figure 2.12: Creating an environment. With *Type* set to *Environment*, you give it a name, folder, host, ports, namespace, and tags (here a dev tag).

Once an environment exists, a connection can link to it by selecting it in the connection’s *Environment* field instead of *manual entry*. The connection then inherits the environment’s host, port, and namespace, so you maintain those details in one place.

Tags

Tags are free-text labels you attach to a connection to enable flexible filtering and search. Unlike environments (which represent a single tier), a connection can carry any number of tags. Examples: *personal*, *client-acme*, *read-only*, *high-memory*.

Tags are displayed inline with the connection name in the list. The Connection Manager provides a tag filter bar at the top of the list: type or select a tag to show only the connections that carry it.

To add or remove tags, open the Edit form for the entry and use the *Tags* field. Type a new tag name and press Enter (or comma) to add it; click the × on an existing tag chip to remove it. Tags are created on first use — there is no separate tag management screen.

Where your connections are stored

ORE Studio keeps your connections — together with their environments, folders, and tags — in a single local SQLite database file

Edit Connection: Local1

Type: Connection

Name: Local1

Folder: Development

Description: Optional description or notes

Environment: ores.dev.local1

Host: localhost

Port: 52001

Namespace: ores.dev.local1

Username: system_admin

Password: Enter new password to change ☐ Show

Tags: +

Test Cancel Save

Figure 2.13: A connection linked to an environment. Selecting `ores.dev.local1` in the *Environment* field populates the host, port and namespace from that environment.

Edit Environment: ores_dev_local1

Type: Environment

Name: ores_dev_local1

Folder: Local1

Description: ORE Studio local development environment 1.

Host: localhost

Port: 42221

HTTP Port: 8080

Namespace: ores.dev.local1

Tags: local1 x local x dev x +

Cancel Save

Figure 2.14: Tags shown as coloured chips in an entry's Edit form (here the `ores_dev_local1` environment, tagged `local1`, `local` and `dev`). Add tags in the *Tags* field, remove one with the `x` on its chip, or click `+` to add another.

named `connections.db`. Your UI settings (preferences and window layout) are stored separately, in the operating system's standard settings store. Neither leaves your machine.

- **Linux:**
 - settings in `~/.config/OreStudio/`;
 - database at `~/.local/share/ores.qt/connections.db`.
- **macOS:**
 - settings in `~/Library/Preferences/`;
 - database at `~/Library/Application Support/ores.qt/connections.db`.
- **Windows:**
 - settings in registry: `HKCU\Software\OreStudio\OreStudio`;
 - database at `%APPDATA%\ores.qt\connections.db`.

If `connections.db` is missing when ORE Studio starts, it creates a new empty one — so the Connection Manager simply opens with no entries.

Backing up and restoring your connections

Because everything you configure here lives in that one `connections.db` file, backing it up is a file copy:

1. Quit ORE Studio, so the database is fully written and not locked.
2. Copy `connections.db` from the location above to a safe place — on Linux, for example:

```
cp ~/.local/share/ores.qt/connections.db ~/connections-backup.db
```

To restore, quit ORE Studio and copy the backup back over `connections.db`. To move your connections to another machine, copy the file to the same location there. The file is self-contained, so a single copy preserves all your connections, environments, folders, and tags.

Connecting and logging in

Once you have at least one connection configured, you can log in.

Selecting the active connection

In the Connection Manager, select the connection you want to use and click **Set as active** (or double-click it). The selected connection becomes the target for the **Connect** button in the main toolbar. The status bar at the bottom of the main window shows the name of the active connection.

You do not have to choose in advance, though: clicking **Connect** opens the login dialog directly, and its *Quick Connect* selector lets you pick the connection there (see [Filtering Quick Connect with labels](#) below).

The login screen

With an active connection selected, click **Connect**. ORE Studio attempts to reach the backend. If the backend is reachable, the *Login* dialog appears.

Figure 2.15: The login dialog. Below the credentials it shows the connection *Label*, a *Quick Connect* selector, and the *Server*, *Port* and *Namespace* the login will target; the footer carries the *Register* link and the build version.

Enter your **username** and **password** and click **Login**. Tick *Remember me* to have ORE Studio recall the username next time. ORE Studio authenticates against the backend’s user database. On success the dialog closes and the main window transitions to the *connected* state: the workspace panels become active, the menu bar is fully enabled, and the status bar shows your username and the connected backend name.

The fields below the credentials describe *which* backend you are logging in to:

- **Label** — the saved connection’s name (e.g. `local1`).
- **Quick Connect** — pick a saved connection to fill *Server*, *Port* and *Namespace* in one step, or leave it on *Enter details manually* to type them yourself.
- **Server, Port, Namespace** — the backend address the login targets.

The *Label* and *Quick Connect* fields work together. The *Label* names the environment — it defaults from the connection (or from the `--instance-name` you launched with); *Quick Connect* then offers the connections for that label, and picking one fills in *Server*, *Port* and *Namespace* so you do not have to type them.

Figure 2.16: The *Label* and *Quick Connect* fields. The *Label* selects the environment; *Quick Connect* offers that environment’s connections and fills in the server details.

If you do not yet have an account, the **Register** link in the footer opens account registration. If authentication fails, an error message is shown below the password field. Check that you are using the correct credentials for this backend; each backend maintains its own user database independently.

Filtering Quick Connect with labels

With only a handful of connections, *Quick Connect* is easy to scan. But a real deployment soon accumulates many — several environments, each with its own system, tenant and per-party logins. Left unfiltered (the *Label* set to *All*), the selector lists every one:

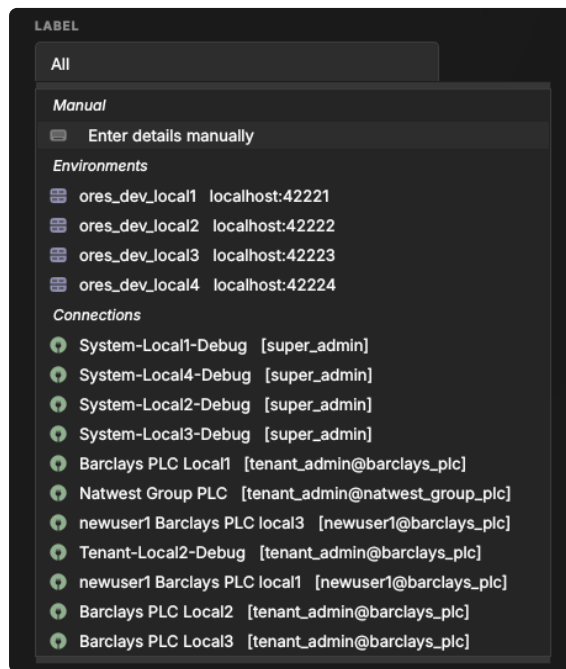


Figure 2.17: Quick Connect with the *Label* set to *All* — every saved connection across all environments and tenants. Hard to pick the right one at a glance.

Each connection carries a *label* — its environment tag, such as *dev*, *local1* or *local2*. The *Label* selector lists those labels; choosing one filters Quick Connect to just the connections that carry it:

With *local1* selected, Quick Connect shows only the *local1* entries — the environment’s own database, its system and tenant logins, and its per-party connections — so you reach the right backend without scrolling past unrelated ones:

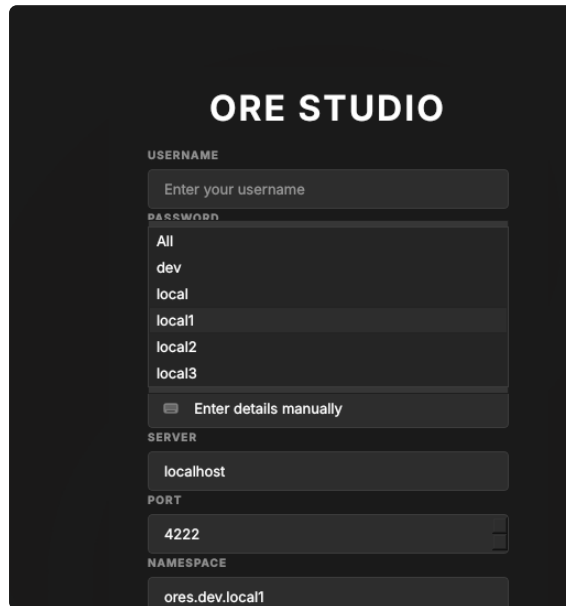


Figure 2.18: The *Label* selector lists the labels found on your connections. Pick one to narrow Quick Connect to that environment.

You normally run one client *per environment*: a client built for `local1` should connect to `local1`, not to staging or production. Rather than pick the label by hand each time, set it when you launch the client — the `--instance-name` command-line option (see [Telling windows apart](#)) opens the client with the *Label* already set, so Quick Connect is pre-filtered to that environment. The `build/scripts/start-client.sh` helper does this for you, taking the label from `ORES_CHECKOUT_LABEL` in your `.env`.

When the connection fails

Not every login failure is a wrong password. ORE Studio talks to the backend over a messaging server (NATS), and if it cannot reach that server — or the services behind it — it tells you which is wrong rather than just reporting "login failed".

If the messaging server itself is unreachable, ORE Studio reports that it cannot connect, along with the host and port it tried:

If the messaging server is running but the application services behind it have not started, ORE Studio says so explicitly — the fix is to start the backend services, not to change your connection:

In both cases your saved connection is fine; correct the backend (start the server or its services) and click **Connect** again.

First login and the default account

On a freshly initialised backend the only account that exists is the default administrator account. Your system administrator will have provided the initial credentials. After first login it is strongly recommended to change the password immediately via *Settings* → *Account*.

Once a tenant has been provisioned (covered in the next chapter, *Initial Setup*), you log in with that tenant's administrator account —

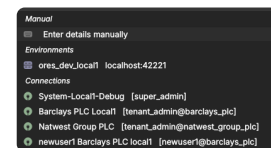


Figure 2.19: Quick Connect filtered to the `local1` label: only that environment's connections remain.

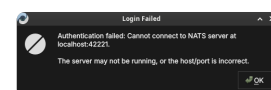


Figure 2.20: A login failure caused by an unreachable messaging server. The host and port ORE Studio tried are shown; the server is likely not running, or the host/port is wrong.

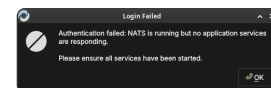


Figure 2.21: A login failure where the messaging server is up but no application services are responding. Ensure the backend services have been started.

the username takes the form `user@tenant_code`. Picking the connection from *Quick Connect* fills in the server, port and namespace for you:

ORE STUDIO

USERNAME

tenant_admin@barclays_plc

PASSWORD

.....

☐ Show password ☐ Remember me

LABEL

local1

QUICK CONNECT

Barclays PLC Local1 [tenant_admin@barc]

SERVER

localhost

PORT

42221

NAMESPACE

ores.dev.local1

Login

Don't have an account? [Register](#)

v0.0.18 local 0736d3f69-dirty

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Figure 2.22: Logging in as a tenant administrator (`tenant_admin@barclays_plc`), with the connection chosen from *Quick Connect*.

Registering a new account

If your deployment allows it, the **Register** link in the login dialog footer opens the *Create Account* form. Fill in a username, email, and password (with confirmation), check the *Server* and *Port* point at your backend, and click **Create Account**. The *Log in* link returns to the login dialog.

Self-registration is not always available: the backend must be set up to permit new users to register themselves. Where it is disabled, accounts are created for you by an administrator instead (see [First login and the default account](#)).

Account registration currently fails with `Account creation failed: NATS connect failed: SSL Error` — the registration path's NATS/TLS connection does not succeed. Until this is fixed, ask an administrator to create your account. The fix is tracked in the story *Fix account registration NATS SSL error*.

Figure 2.23: The Create Account form, opened from the *Register* link in the login dialog footer.

Connection status and reconnection

The status bar always shows whether you are connected. After a deliberate log-out — or once a stopped backend is running again — use the **Connect** button to re-establish the session: your saved connection is retained, so you only re-enter your password.

ORE Studio does not yet reliably detect a backend that drops *while you are logged in*. If the messaging server (NATS) or the services behind it are stopped mid-session, the client may keep looking connected until your next action fails or hangs. Detecting the drop and showing a clear connection-lost state is planned work, tracked in the story *Detect and report NATS disconnection*.

The status bar

The status bar runs along the bottom of the main window and provides a continuous read on the application's connection state. It is always visible regardless of what is open in the workspace.



The status bar is divided into zones from left to right:

- **Connection name** — the name of the active connection as entered in the Connection Manager (e.g. "Local dev"). Clicking this zone opens the Connection Manager directly.

Figure 2.24: The status bar in the fully connected, logged-in state: from left to right, the environment marker (`local1`), the username (`super_admin`), the server (`localhost:42221`), the active party (Root / System Party), and the green connected indicator.

- **Environment badge** — the coloured environment label (e.g. Production in red) if one is assigned to the active connection. Absent when no environment is set. This is the most prominent safety indicator: always glance at the environment badge before performing any write operation.
- **Username** — the account name of the currently logged-in user. Absent in the disconnected state.
- **Server** — the backend address (host and port) the session is connected to, e.g. localhost:42221.
- **Active party** — the party context for the session (e.g. Root / System Party), set at login when your account spans multiple parties.
- **Connection indicator** — a coloured icon at the right showing the live link state (green when connected).

The status bar changes appearance to reflect the connection state:

- **Disconnected** (no active connection) — grey; shows "Not connected".
- **Connecting** (handshake in progress) — animated indicator; shows "Connecting to /name/...".
- **Connected and logged in** — normal; shows all zones as described above.

A distinct *connection lost* state for a backend that drops mid-session is planned but not yet implemented — see [Connection status and reconnection](#) above.

In the disconnected state the strip is reduced to the environment marker and the connection label, with the broken-link icon on the right:

The connected state is shown at the start of this section. Compare it with the disconnected strip above to recognise at a glance which state the application is in.

Starting ORE Studio from the command line

ORE Studio can be launched directly from a terminal, which is useful for scripting, for telling several windows apart, or for pointing the application at a non-default configuration directory. Run `ores --help` to see the full list of accepted options for your installed version; the most commonly used ones are documented below.

```
ores --help
```

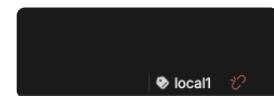


Figure 2.25: The status bar in the disconnected state, showing the environment marker, the connection label (local1), and the broken-link indicator.

Telling windows apart (instance colour and name)

When you run more than one ORE Studio window at once — for example against different backends, or from separate checkouts — you can give each window a distinct identity so you can tell them apart at a glance. This is *not* a light/dark theme; it is purely a per-window marker.

- `--instance-color` HEX draws a small coloured circle in the status bar in that colour (a 6-digit RGB hex, e.g. F44336 for red). Give each window a different colour.
- `--instance-name` NAME (short form `-n`) labels the window with a name, also shown in the status bar, so you can see which window is which.

```
ores --instance-name "Local dev" --instance-color 2196F3
```

The colour and the name are independent: the colour is only a visual marker, and the name is what identifies the instance. If you launch via `build/scripts/start-client.sh`, its `--colour red|green|blue|<hex>` sets the marker colour and `--name` sets the instance name; when you omit `--name`, the name comes from `ORES_CHECKOUT_LABEL` in your `.env` — never from the colour.

Configuration directory

By default ORE Studio stores its UI settings — preferences and window layout — in the operating system's standard settings store. The exact per-OS locations are listed under [Where your connections are stored](#); your saved connections live separately in `connections.db`. To use a different directory:

```
ores --config-dir /path/to/config
```

This is useful when running multiple isolated instances, or when keeping configuration under version control for team-shared settings.

Selecting a connection at startup

To bypass the Connection Manager and connect immediately to a named connection:

```
ores --connection "Local dev"
```

ORE Studio will start, set the named connection as active, and open the login dialog automatically. The connection name must match exactly (case-sensitive) a saved entry in the Connection Manager.

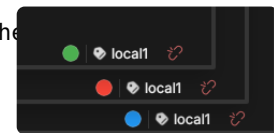


Figure 2.26: The status bars of three windows running the same `local1` instance, each given a different `--instance-color` (green, red, blue) so you can tell them apart at a glance.

Logging and diagnostics

For troubleshooting, verbosity can be increased:

```
ores --log-level debug    # verbose output to stdout
ores --log-file /tmp/ores.log # write log to a file instead
```

Log output includes connection lifecycle events, authentication attempts, and backend protocol messages. The debug level is intended for issue reporting and development; it produces high-volume output and should not be left enabled during normal use.

Finding the application version

Knowing exactly which build you are running matters when reporting an issue or checking client/backend compatibility. ORE Studio shows its version in several places.

The title bar of the main window always shows the version next to the application name:

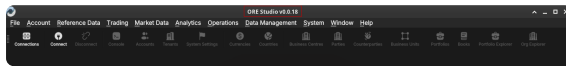


Figure 2.27: The version in the main window title bar, alongside the full menu bar and toolbar of the connected application.

It also appears in the lower-right corner of the splash screen at start-up:



Figure 2.28: The build version in the lower-right corner of the splash screen.

...and in the footer of the login dialog, below the *Register* link:

From the command line, `ores --version` prints the client version string (e.g. ORE Studio 0.0.19) and exits — handy in scripts or when filing a bug report.

```
ores --version
```

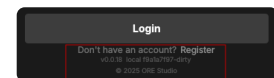


Figure 2.29: The version string in the login dialog footer, beneath the *Register* link.

Running in the background and the system tray

While ORE Studio is running it places an icon in your desktop's system tray (notification area). The icon keeps the application reachable when its window is minimised or hidden, and shows the instance name so you can tell multiple windows apart.

Logging out

To end your session, choose *File* → *Log Out* or click the **Disconnect** toolbar button. ORE Studio closes the connection to the backend and



Figure 2.30: The ORE Studio icon in the system tray, labelled with the instance name so you can identify the window it belongs to.

returns to the disconnected main window. Your saved connection configurations are preserved; you can log back in at any time.

Logging out does not stop the backend service — it only terminates your client session. Other users connected to the same backend are unaffected.

To close the application entirely, quit the window (or choose *File* → *Exit*). ORE Studio asks you to confirm so you do not lose an active session by accident:

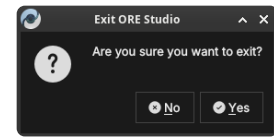


Figure 2.31: The exit confirmation. Click *Yes* to close ORE Studio, or *No* to keep working.

Initial Setup

Understanding the multi-tenant, multi-party model

Before you can use ORE Studio for the first time, the system must be provisioned — meaning the initial administrative structure must be established. Understanding why this step exists requires a brief introduction to how ORE Studio organises data.

Tenants

A *tenant* is a fully isolated organisational space within an ORE Studio deployment. Each tenant has its own users, its own reference data (currencies, counterparties, books), and its own analytics results. Data belonging to one tenant is completely invisible to another tenant; there is no cross-tenant data leakage by design.

A typical deployment has one tenant per organisation. If you are running ORE Studio for your own firm, you will create one tenant representing your organisation. A managed-service provider running ORE Studio on behalf of multiple clients might create one tenant per client.

ORE Studio supports four tenant types, each with different operational controls:

- **System** — platform administration; one per deployment; platform-managed.
- **Production** — real customer organisations with live data; strict controls.
- **Evaluation** — demos, UAT, training, and exploratory testing; relaxed controls.
- **Automation** — programmatic test infrastructure; not for human use; no controls.

The **system tenant** is special — it is created automatically during system provisioning and cannot be deleted. It is the home of the platform administrator accounts. There is exactly one system tenant per deployment. All other tenants are created by platform administrators working from the system tenant.

Production tenants are for live operations. They enforce strict controls including four-eyes authorisation for sensitive operations and KYC-gated counterparty onboarding.

Evaluation tenants provide a realistic but relaxed environment for demonstrations, user acceptance testing, and training. Bulk data import from external sources (such as the GLEIF/LEI registry) is available in evaluation tenants but not in production tenants.

Parties

Within a tenant, a *party* is a business unit — a legal entity, a branch, a trading desk, or any other organisational subdivision. Parties form a hierarchy: a parent party can see all data belonging to its children and grandchildren; a child party sees only its own data and that of its own descendants.

Every tenant has exactly one **system party**, created automatically when the tenant is provisioned. The system party is the administrative home for tenant administrator accounts. It is not a business entity and should not be used for trading activity.

Business parties — the entities that actually own trades, books, and analytics results — are **operational parties**. You create these after the tenant is provisioned, using the Party Provisioner.

In financial industry usage, the set of operational parties representing your own organisation is called *the house*. The house hierarchy models your corporate structure: the root party is your top-level legal entity; its children are subsidiaries, branches, or regional offices; their children are individual trading desks or booking centres. Trades, positions, and risk reports all belong to a specific party within the house.

Distinct from the house are *counterparties* — the external legal entities your house trades with (banks, broker-dealers, corporates, funds). Counterparties are reference data: they are attached to trade tickets to identify the facing entity, but they do not own books or belong to the party hierarchy. The house versus counterparty distinction appears throughout OreStudio's UI and data model; keeping it clear is important when working with trades and risk reports.

A user logs in to a specific party within the house. Their data visibility is determined by their position in the hierarchy: a user logged in at "Acme Group" sees data from both "Acme London" and "Acme New York"; a user logged in at "Acme London" sees only their own data.

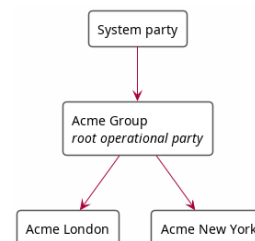


Figure 3.1: A simple party hierarchy showing the house: the system party at the top, the root operational party (Acme Group) below it, and two child operational parties (Acme London, Acme New York).

The provisioning sequence

A freshly installed ORE Studio backend goes through a fixed sequence before it is ready for normal use. Each step requires a specific login identity:

1. **System provisioning** — no login is needed; the backend is in *provisioning mode* and the System Provisioner wizard launches automatically the first time any client connects. This step creates the platform administrator account and provisions the first tenant and its tenant administrator. Until it completes, no normal user logins are accepted.
2. **Tenant provisioning** — log out of the system administrator session (or simply connect as a new user) and **log in as the tenant administrator** created in step 1. OreStudio detects that this account has never logged in before and launches the Tenant Provisioner automatically. This step seeds the tenant with reference data and creates the house party hierarchy.
3. **Party provisioning** — **log in to a specific house party** using the tenant administrator account, or any operational account that holds party setup permissions for that party. OreStudio detects that the party has not yet been provisioned and launches the Party Setup wizard. Repeat this step for each party in the hierarchy that needs its own counterparties, books, and reports.

Each step has a corresponding wizard in the ORE Studio UI. The following sections walk through each one.

The System Provisioner wizard

The first time you connect to a backend that has never been initialised, ORE Studio detects *provisioning mode* and launches the **New System Provisioner** automatically. This one-time wizard creates the platform administrator account, chooses a single- or multi-tenant layout, and provisions the first tenant together with its own administrator.

Welcome

The welcome page confirms the system is in provisioning mode and outlines the steps: create the administrator account, choose the setup mode, and create the first tenant. Click **Next**.

Create the administrator account

Enter a **username**, **email**, and **password** (with confirmation) for the platform administrator. Keep these credentials safe — this account has unrestricted access to the entire deployment. Click **Create & Continue**.

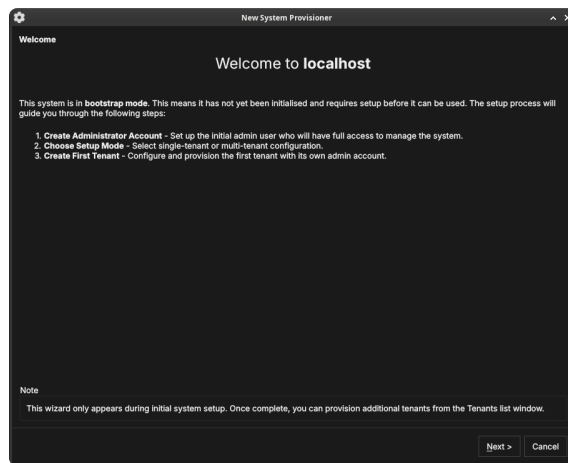


Figure 3.2: The System Provisioner welcome page, shown when the backend is in provisioning mode. It lists the three setup steps.

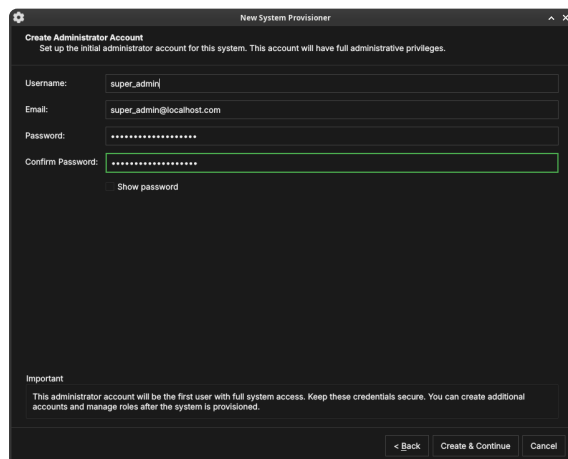


Figure 3.3: Creating the platform administrator account — the first account, with full administrative privileges over the whole deployment.

Choose the setup mode

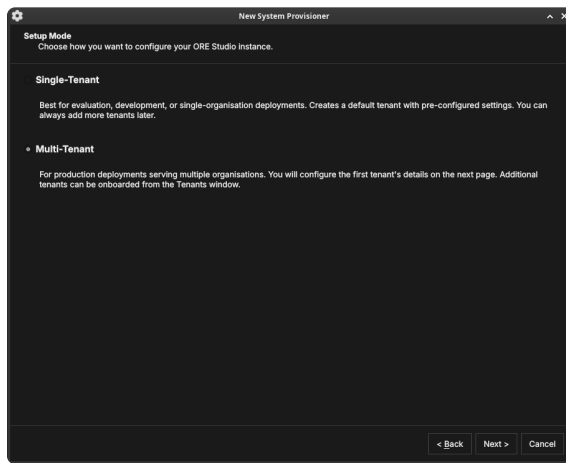


Figure 3.4: The Setup Mode page. *Single-Tenant* creates a default tenant with pre-configured settings; *Multi-Tenant* lets you configure the first tenant in detail and onboard more later.

Choose how the deployment is organised:

- **Single-Tenant** — best for evaluation, development, or a single organisation. Creates a default tenant with sensible defaults; you can add more tenants later.
- **Multi-Tenant** — for production deployments serving several organisations. You configure the first tenant's details on the next page and onboard additional tenants from the Tenants window.

Click **Next**.

Tenant details

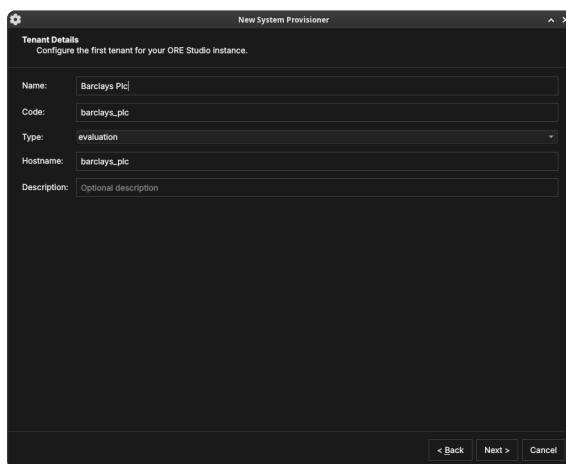


Figure 3.5: The first tenant's details: a display name, a short code, a tenant type, and a hostname.

Configure the first tenant:

- **Name** — the organisation's display name (e.g. "Barclays Plc").
- **Code** — a short machine code used internally (e.g. `barclays_plc`).
- **Type** — the tenant type (see [Tenants](#) above): *System*, *Production*, *Evaluation*, or *Automation*.

- **Hostname** — the host identifier for the tenant.

Click **Next**.

Tenant administrator account

Figure 3.6: The tenant administrator account. This account administers the new tenant and is independent of the platform administrator.

Create the initial administrator for the new tenant — a **username**, **email**, and **password**. This account is given the TenantAdmin role: full control within the tenant, but no cross-tenant access. Click **Provision Tenant**.

Provisioning

Figure 3.7: The provisioning step. ORE Studio creates the tenant and its administrator, logging progress as it goes.

ORE Studio provisions the tenant and creates its administrator account, logging each step. When it finishes, click **Continue**.

Setup complete

The final page summarises what was created and tells you the next step: log in as the tenant administrator to run the Tenant Provisioner. To create further tenants later, use *System* → *Identity* → *Tenants*

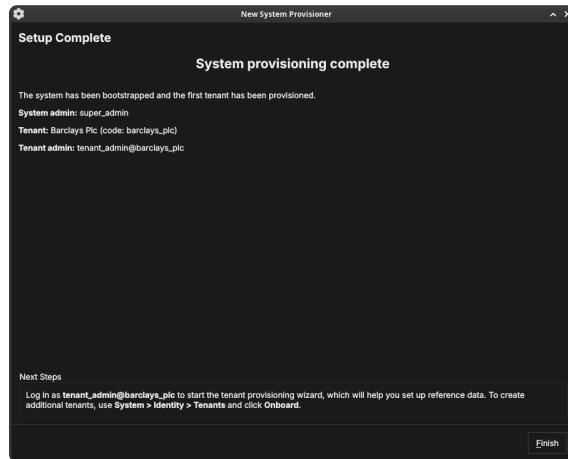


Figure 3.8: The System Provisioner summary: the platform admin, the first tenant, and the tenant admin that were created, with next-step guidance.

and click **Onboard**. Click **Finish** — the backend leaves provisioning mode and the login dialog appears.

The Tenant Provisioner wizard

The first time you log in as a tenant administrator, ORE Studio launches the **New Tenant Provisioner**. It seeds the tenant with reference data and, optionally, an initial party hierarchy. You can skip it with **Cancel** and set everything up by hand later from the Data Librarian and Parties windows.

Welcome

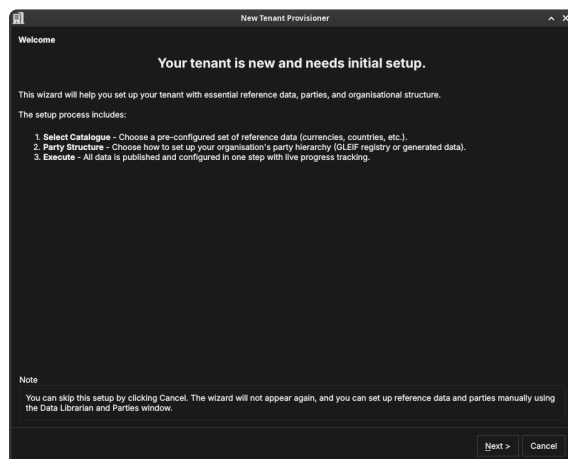


Figure 3.9: The Tenant Provisioner welcome page, listing its three steps: select a reference-data catalogue, choose a party structure, and execute.

The welcome page outlines the steps: select a catalogue, choose how to populate the party hierarchy, and execute. Click **Next**.

Select a catalogue

A *catalogue* is a ready-made bundle of reference data. The *Base System* catalogue provides industry-standard ISO and FpML data — country

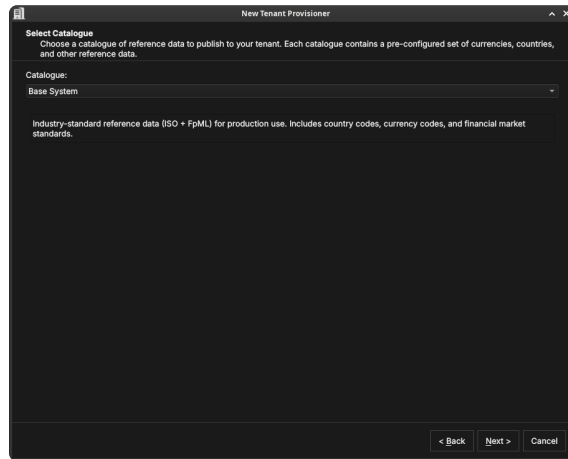


Figure 3.10: Selecting a reference-data catalogue — a pre-configured set of currencies, countries, and market standards.

codes, currency codes, and financial-market standards — suitable for production use. Choose a catalogue and click **Next**.

Choose a data source

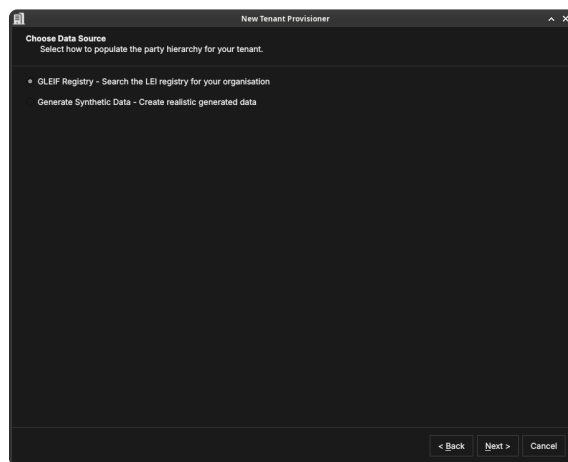


Figure 3.11: Choosing how the tenant’s internal party hierarchy — your organisation’s own legal entities and business units — will be seeded.

This step seeds the *internal* party hierarchy: the legal entities and business units that represent your own organisation — what practitioners call *the house*. As described in [Parties](#) above, these are the operational parties that own trades, books, and analytics results.

The data source choice does **not** affect counterparties (the external entities your organisation trades with — those are imported separately in the Party Setup wizard). It only determines how the house structure is created.

In a group context — for example a bank holding company with separate London and New York subsidiaries — the house hierarchy reflects the corporate structure. The root party is the top-level legal entity; its children are the subsidiary entities; their children are trading desks, books, or booking centres. Each node in this tree represents a real organisational unit, and users log in to the node that corresponds to their own entity.

Two seeding methods are available:

- **GLEIF Registry** — searches the global LEI registry for your real organisation. The entity you select becomes the root party, and its corporate descendants in the GLEIF hierarchy — subsidiaries, branches, and related entities — are created automatically as child parties. This is the recommended approach for production deployments: the GLEIF data provides real legal entity names, LEI codes, and verified corporate relationships.
- **Generate Synthetic Data** — creates a realistic but fictional party hierarchy using generated names. Use this for evaluation tenants, demonstrations, or testing where you want a plausible structure without importing real organisational data.

Click **Next**.

Party setup (optional)

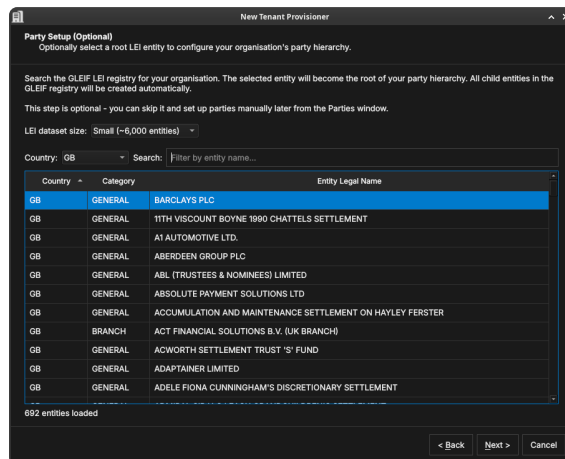


Figure 3.12: The optional Party Setup step. Search the GLEIF LEI registry for your root entity (here Barclays PLC); its corporate descendants are created as child parties automatically.

It helps to be clear about what this step does and does not do.

What it creates: the party *nodes* — the named entries in the hierarchy that represent your legal entities and business units. Each node gets a name, a LEI code, and a position in the tree. After this step OreStudio knows that "Barclays PLC" exists as a party, that it is the parent of "Barclays Bank UK PLC", and so on. The hierarchy structure is in place.

What it does not create: the operational *content* within each party — the books, portfolios, business units, counterparties, and report definitions that each party uses day-to-day. That content is added separately, after provisioning, by the Party Setup wizard (described in [The Party Provisioner wizard](#) below). Think of this step as building the org chart; the Party Setup wizard then stocks each office.

To use the GLEIF registry, pick an **LEI dataset size** (which controls how much of the registry is loaded for searching), filter by **Country** if needed, and search by entity name. The entity you select becomes the *root house party* — the top of your internal hierarchy. Its corporate children and grandchildren in the GLEIF registry (subsidiaries,

branches, booking centres) are created automatically as child operational parties. You do not need to build the structure by hand; the GLEIF corporate hierarchy data does it for you.

This step is optional — skip it to create parties manually later from the Parties window. Click **Next**.

Publishing

Setting Up Tenant
Publishing reference data and configuring your organisation. Please wait — this may take several minutes.

Publishing datasets... (31 dispatched)
Step 28 of 28 in progress...

#	Name	Status	Warnings	Started At	Completed At
21	fpml.party_relationship_type	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
22	fpml.party_role	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
23	fpml.party_role_type	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
24	fpml.person_role	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
25	fpml.regulatory_corporate_sector	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
26	fpml.reporting_regime	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
27	fpml.supervisory_body	completed	—	2026-05-25 23:27:44Z	2026-05-25 23:27:44Z
28	gleif.lei_counterparties.small	in progress	—	2026-05-25 23:27:44Z	—
29	—	pending	—	—	—
30	—	pending	—	—	—
31	—	pending	—	—	—

Publishing catalogue 'Base System' with GLEIF root LEI: BARCLAYS PLC (dataset: small)
Catalogue workflow started: 31 datasets dispatched.

< Back Commit Cancel

Figure 3.13: The publishing step, showing 31 datasets being dispatched in dependency order. Dataset 28 (gleif.lei_counterparties.small) is in progress; datasets 29–31 are pending.

OreStudio runs a multi-step publishing pipeline to populate the tenant with all the data selected in the previous steps. The pipeline dispatches datasets in the correct dependency order — classification tables before the entities that reference them, catalogue data before GLEIF data — and shows live progress in the table.

Each row in the table is one dataset. The columns are:

- **#** — the dataset’s position in the dependency graph (zero-based). Datasets with lower numbers are prerequisites for those with higher numbers.
- **Name** — the dataset identifier, in `source.dataset` form. The prefix identifies the data source: `iso.*` datasets contain ISO standard data (currencies, countries); `fpml.*` datasets contain FpML reference classifications (party types, roles, regulatory sectors); `gleif.*` datasets contain GLEIF LEI entity data (counterparties, house parties).
- **Status** — `pending` (not yet started), `in_progress` (actively loading), or `completed` (finished successfully). A red status indicates a failure.
- **Warnings** — any non-fatal issues encountered during that dataset’s load, displayed as a count. Click the row to see details.
- **Started At / Completed At** — UTC timestamps for the dataset’s execution window.

The log panel at the bottom provides a running narrative: which catalogue is being published, how many datasets were dispatched,

and the final outcome for each phase (reference data, organisation, activation).

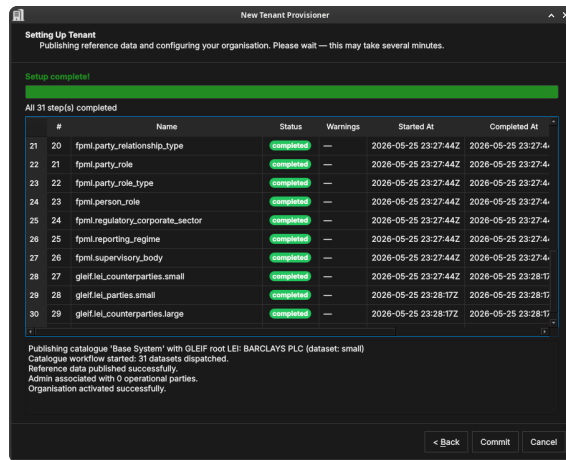


Figure 3.14: All 31 datasets completed. The log confirms: reference data published, organisation associated with parties, and the tenant activated.

When all rows show completed and the log ends with "Organisation activated successfully", click **Commit** to finalise the setup, or **Back** to change your selections.

Setup complete

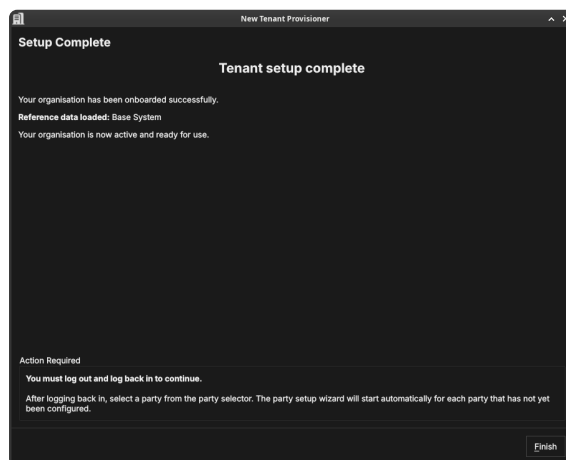


Figure 3.15: The Tenant Provisioner summary, confirming the reference data and parties that were set up.

The final page confirms what was created. Click **Finish**. The tenant now has its reference data and party hierarchy in place.

The Party Provisioner wizard

Where the Tenant Provisioner establishes the party *hierarchy*, the **Party Setup** wizard configures the operational structure *within* a party: it imports counterparties and creates a standard set of risk-report definitions. It runs for a party that needs initial setup, and can be skipped with **Cancel** (counterparties and reports can be configured later from the application menus).

Welcome

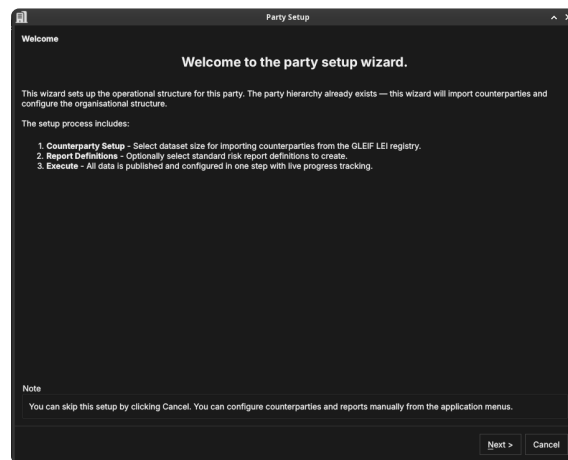


Figure 3.16: The Party Setup welcome page, listing its steps: counterparty import, report definitions, and execute.

By this point the house hierarchy — the party nodes representing your organisation’s legal entities — already exists. This wizard populates one specific party within that hierarchy: it attaches the external counterparties that this party trades with, creates the internal organisational structure (business units, portfolios, and books), and defines the risk reports that will run on its data. Click **Next**.

Counterparty import

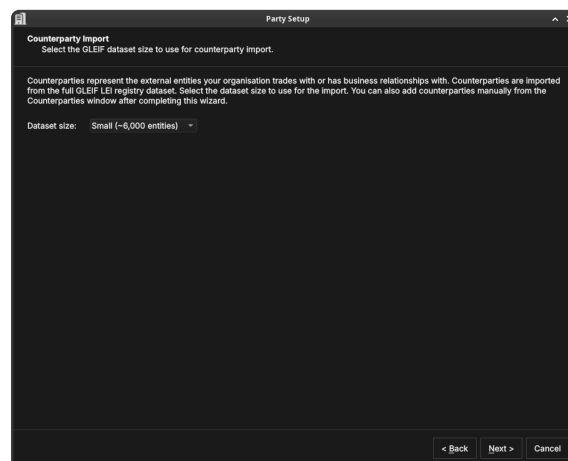


Figure 3.17: Choosing the GLEIF dataset size for importing counterparties. The dataset size controls how many real-world legal entities are loaded from GLEIF as counterparties.

Counterparties are the external legal entities your organisation trades with — banks, broker-dealers, corporates, funds. Where parties (the house) represent your own organisational structure, counterparties represent the other side of your trades. The distinction is important: house parties own books and positions; counterparties appear on trade tickets as the facing entity.

OreStudio imports counterparties from the GLEIF LEI registry. This is a deliberate design choice: GLEIF contains over two million real legal entities with verified names, LEI codes, countries of incorporation, and corporate relationships. Importing from GLEIF means

your OreStudio instance starts with a realistic, authoritative counterparty population rather than a hand-crafted list. In an evaluation or demonstration tenant this is especially valuable — trades booked against real counterparty names (Deutsche Bank, JP Morgan, Black-Rock) look and behave exactly as they would in production, making scenario testing meaningful.

The **dataset size** controls how many GLEIF entities are loaded: smaller sizes load a representative subset quickly; larger sizes import a more complete global population. You can add individual counterparties later from the Counterparties window regardless of which size you choose. Click **Next**.

Report definitions

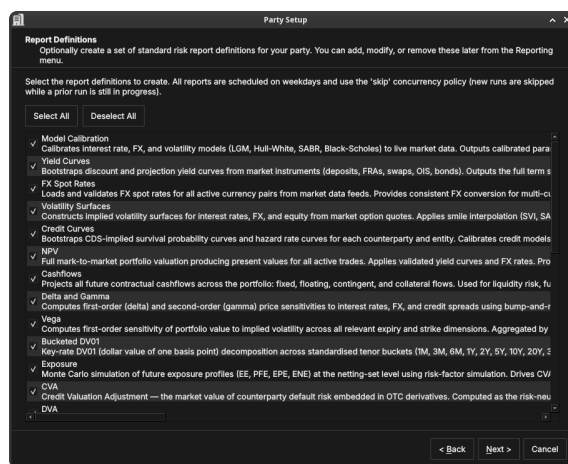


Figure 3.18: Selecting standard risk-report definitions to create — NPV, sensitivities, exposure, CVA, and more. All are scheduled on weekdays.

Optionally create a set of standard risk-report definitions — model calibration, yield curves, FX spot rates, volatility surfaces, credit curves, NPV, cashflows, Delta and Gamma, Vega, bucketed DV₀₁, exposure, CVA, DVA, and others. Use **Select All** / **Deselect All**, or tick individual reports; they can be edited later from the Reporting menu. Click **Next**.

Execute

OreStudio runs the party setup in four phases, shown in the log panel:

1. **Phase 1: importing counterparties** — loads the selected GLEIF counterparty dataset into the party's counterparty table. The progress table shows one row per dataset (e.g. `gleif.lei_counterparties.small`), with the same Status / Started At / Completed At columns as the Tenant Provisioner publishing step.
2. **Phase 2: publishing organisation structure** — creates the internal organisational entities within the party: business units (`testdata.business_units`), portfolios (`testdata.portfolios`), and books (`testdata.books`). Books are the lowest-level containers that own individual trades and positions.

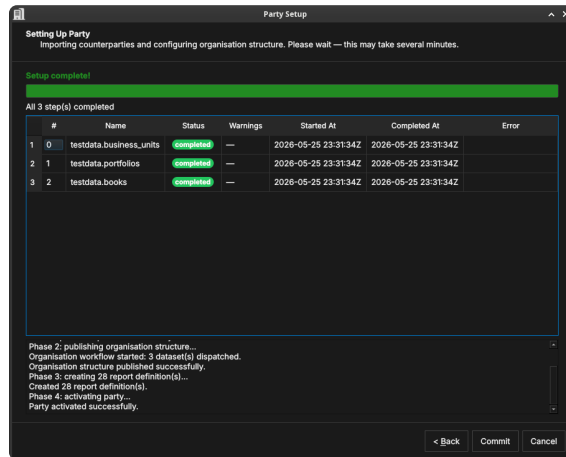


Figure 3.19: The execute step showing three datasets completed: testdata.business_units, testdata.portfolios, and testdata.books. The log below shows the four phases completed in sequence.

3. **Phase 3: creating report definitions** — creates the scheduled risk-report configurations selected in the previous step.
4. **Phase 4: activating party** — marks the party as operational so users can log in to it and begin booking trades.

When the log ends with "Party activated successfully", click **Commit**.

Setup complete

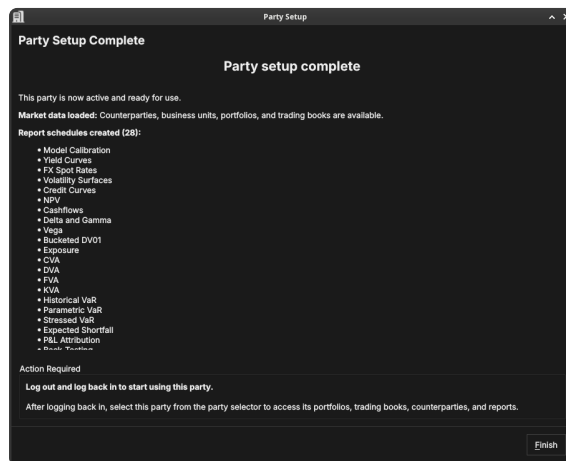


Figure 3.20: The Party Setup summary, confirming the counterparties, structure, and reports that were created.

The final page confirms what was set up. The party is now fully operational: it has counterparties, an organisational structure, and scheduled risk reports.

Choosing a party at login

Some accounts are associated with more than one party — for example a group administrator who can act for several entities. When you log in with such an account, ORE Studio asks which party context to use for the session.

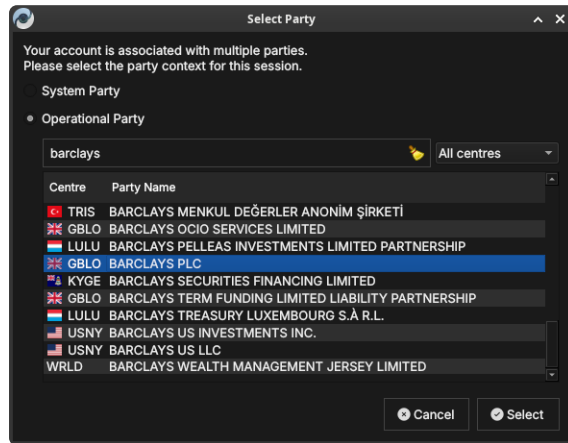


Figure 3.21: The Select Party dialog. Choose the System Party or an Operational Party; filter by booking centre or search by name.

Choose **System Party** for administrative work, or **Operational Party** to work within a specific business entity. Filter the list by booking **centre**, or type in the search box to narrow it.

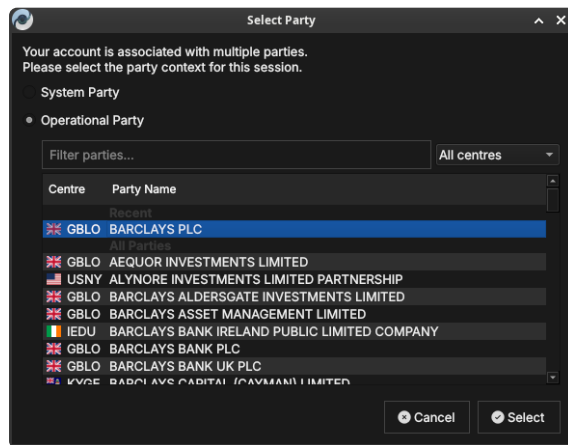


Figure 3.22: The Select Party dialog with a recently used party listed under *Recent* at the top for quick access.

Select a party and click **Select**. Your data visibility for the session is determined by the party you choose and its position in the hierarchy.

Part II

Reference Data

Reference data is the slowly-changing master data that underpins all financial activity in OreStudio — the currencies, counterparties, books, and market conventions that trades and analytics refer to. This part documents each reference data entity: what it represents, how to manage it through the Qt interface, and how to work with it from the shell and CLI.

Reference Data

Overview

This chapter introduces OreStudio's reference data model. It explains what reference data is and how it differs from market data and trades, then covers the data quality framework that governs it: the six industry-standard DQ dimensions, the bitemporal storage model, versioning and immutable history, provenance tracking, and the structured change reason system. Readers who want to jump straight to managing specific entity types can skim this chapter and return to it when questions arise about audit trails or data quality controls.

What is Reference Data?

REFERENCE DATA is the static or slowly-changing master data that all financial activity depends on. It defines the vocabulary of the system — the currencies a trade is denominated in, the counterparty on the other side of a deal, the book it settles into, the country a legal entity is incorporated in. Without it, nothing else can be described precisely.

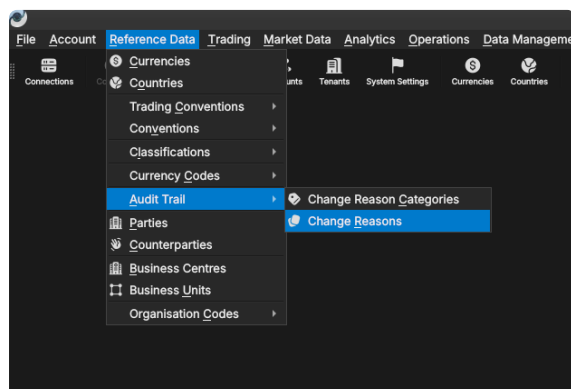


Figure 4.1: The Reference Data menu showing the full set of entity types managed by OreStudio. The Audit Trail sub-menu provides access to Change Reasons and Change Reason Categories.

OreStudio's reference data covers a wide range of entity types:

- **Currencies** — ISO 4217 codes, display formatting, and rounding rules.
- **Countries** — ISO 3166 alpha-2 codes and names.
- **Trading conventions and calendars** — holiday calendars, settlement day rules, date rolling conventions, and tenor definitions that drive date arithmetic in trade processing and valuation.
- **Counterparties** — the external legal entities your organisation trades with.
- **Books** — the internal trading books that own positions.
- **Business units and portfolios** — the organisational structure within a party.
- **Party types and statuses** — the classification vocabulary for parties.
- **Classifications** — supplementary taxonomies such as currency market tiers and monetary natures.

Reference data is distinct from the other two main categories of data in the system:

- **Market data** — prices, FX rates, yield curves, and volatility surfaces that update continuously throughout the trading day. Market data is time-stamped and immutable once recorded; a new observation does not overwrite an old one.
- **Trades** — specific financial transactions between parties with defined cashflows, valuation models, and lifecycle states. Trades reference the reference data (they need a currency, a counterparty, a book) but they are not reference data themselves.

The key property of reference data is that it changes slowly and deliberately. A currency's rounding convention is not something that shifts intraday — it is corrected through a controlled, audited process. OreStudio enforces this with a comprehensive data quality framework described in this chapter.

GLEIF and the Legal Entity Identifier (LEI)

The Global Legal Entity Identifier Foundation ([GLEIF](#)) is a not-for-profit organisation that oversees the global LEI system on behalf of financial regulators. A **Legal Entity Identifier (LEI)** is a 20-character alphanumeric code that uniquely identifies a legal entity participating in financial markets — a bank, a corporation, a fund, a branch. LEIs are mandated by major regulatory frameworks including MiFID II, EMIR, Dodd-Frank, and Basel III for trade reporting and counterparty identification.

GLEIF publishes the full registry of LEI entities and their corporate hierarchies (parent-child relationships) as open data. OreStudio uses

the GLEIF dataset to seed counterparties and, when a party's root LEI is selected during tenant provisioning, to populate the initial party hierarchy from real organisational data.

A **BIC** (Bank Identifier Code, ISO 9362) is an 8 or 11 character code identifying a specific financial institution, used in SWIFT messaging for settlement routing. GLEIF publishes a LEI-to-BIC mapping dataset that OreStudio also imports, allowing settlement systems to resolve a counterparty's BIC from its LEI.

Data Quality

Financial calculations are only as reliable as the data they consume. A misspelled currency name is a cosmetic nuisance; an incorrect rounding rule, a wrong counterparty identifier, or a stale market tier classification can silently propagate into risk figures, margin calls, collateral calculations, and regulatory reports. The consequences range from minor reconciliation breaks to significant financial loss or regulatory censure.

OreStudio's reference data layer is designed around the definition of data quality from ISO 8000 and the DAMA Data Management Body of Knowledge (DAMA-DMBOK): *data quality is the measure of how well a dataset satisfies the requirements of its intended business use*. In financial markets this means data must be not only accurate but also complete, consistent, timely, valid, and unique — the six industry-standard DQ dimensions described below.

The Six Dimensions

Accuracy is the degree to which data values agree with their authoritative golden source. OreStudio seeds currencies from ISO 4217, countries from ISO 3166, and counterparty identifiers from the GLEIF LEI registry. Where a discrepancy arises between a record in OreStudio and its source, the source takes precedence; the system provides tooling to re-import from authoritative datasets.

Completeness means all mandatory attributes are present. The service layer rejects saves that omit required fields and returns a validation error; the Qt UI marks missing mandatory fields visually before a save is attempted.

Consistency is the absence of contradictions between related entities. A currency's rounding type must be drawn from the governed rounding-types table; its market tier from the currency-market-tiers table. Foreign-key constraints in the database enforce this at the persistence layer.

Timeliness means data is available when calculations need it. The tenant provisioner imports standard catalogues at setup time so foundational reference data is ready before any trading activity begins. NATS events propagate updates to all connected clients in real time.

Validity is adherence to business rules and formats. ISO code

lengths, numeric code ranges, currency format strings, and rounding precision bounds are all validated at save time.

Uniqueness ensures no duplicate records exist for the same entity. Primary key constraints at the database level prevent duplicates; the service layer surfaces a clear error if a duplicate is attempted.

Bitemporality

The most important architectural property of OreStudio's reference data layer is its use of *bitemporality* — the recording of two independent time dimensions for every stored fact.

The first dimension is **valid time**, recorded in `valid_from` and `valid_to` columns on every row. Together they form an open interval `[valid_from, valid_to)` bounding the period during which this version of the record is authoritative. The live (current) row carries `valid_to = 9999-12-31 23:59:59` — a sentinel meaning "no known expiry". When an update is saved, the database trigger closes the existing row by setting `valid_to` to the current time, and inserts a new row with `valid_from = now` and `valid_to = sentinel`. The application never writes `valid_from` or `valid_to` directly — the trigger owns both columns.

The second dimension is **transaction time**, surfaced to the application as the `recorded_at` field. It records when this version was written to the database — independently of what period the version is valid for. Transaction time answers the question "what did the system believe at a given moment?"

All timestamps in OreStudio are stored and processed as UTC throughout. The Qt UI converts timestamps to your local timezone for display only; every timestamp you see in the application is a local-time rendering of a UTC value stored in the database.

The combination of the two time dimensions allows OreStudio to answer questions that neither alone could answer:

- *What is the current record for USD?* — the live row has `valid_to = sentinel`.
- *What did the USD record look like on a specific past date t ?* — find the row where `valid_from < t < valid_to`.
- *When did we first record the current name for a currency?* — inspect the `recorded_at` of the version where the name field changed.
- *Revert to an earlier version* — inserts a new row copied from the historical version, leaving all intermediate versions intact.

Bitemporality is particularly important for **calculation reproducibility**. Being able to reconstruct the exact state of all reference data as it existed on a past valuation date is a prerequisite for explaining historical calculation results, for regulatory back-testing, and for resolving disputes about previously reported figures.

Versioning and Immutable History

Every save to a reference data entity creates a new, numbered version. The version counter starts at 1 on the first save and increments monotonically. History is immutable: no version is ever deleted or modified. Even a "revert" operation creates a new version — it does not roll the counter back or remove intermediate history.

This immutability guarantee is deliberate. Compliance frameworks such as MiFID II and EMIR require that firms demonstrate the state of their data at any past point in time. OreStudio's history tables provide this proof unconditionally.

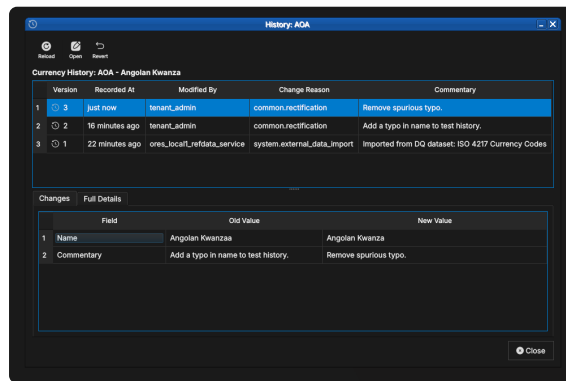


Figure 4.2: The Currency History dialog for the Angolan Kwanza, showing three versions. The Changes tab shows a field-by-field diff for the selected version against its predecessor.

The history dialog, accessible from every detail dialog's title bar or the list window's right-click menu, shows the full version list. Selecting a version populates a detail panel. The **Changes** tab shows a field-by-field diff; the **Full Details** tab shows the complete record at that version. The **Revert** button reinstates a historical version as a new current version.

Provenance

Every version of every reference data record carries six standard provenance fields. The **Provenance** tab in every detail dialog shows them read-only; it is disabled in create mode since no provenance exists before the first save.

1. version — Version

Set by DB trigger
UI label Version

Monotonically increasing integer starting at 1, incremented by the database trigger on every save. Never written by the application.

2. modified_by — Modified By

Set by Application
UI label Modified By

The username of the account that submitted the save request.

3. `performed_by` — Performed By

Set by Application
 UI label Performed By

The username on whose behalf the change was performed. In automated workflows, a service account may submit a change on behalf of a human operator — in that case `modified_by` identifies the service and `performed_by` identifies the human, preserving accountability at both layers.

4. `recorded_at` — Recorded At

Set by DB trigger
 UI label Recorded At

The UTC wall-clock timestamp at the moment the row was written, set by the database trigger. Like all timestamps in OreStudio it is stored as UTC and displayed in your local timezone.

5. `change_reason_code` — Change Reason

Set by Application
 UI label Change Reason

A structured code drawn from the change reasons table identifying the business justification for the change. See [Change Reasons](#) below.

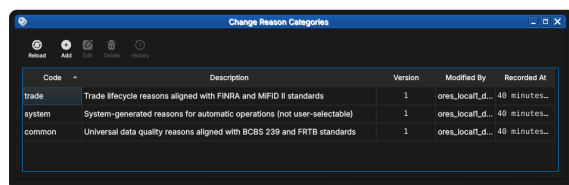
6. `change_commentary` — Commentary

Set by Application
 UI label Commentary

A free-text note accompanying the change. Mandatory for some reason codes, optional for others. Displayed in the history dialog and stored permanently with the version.

Change Reasons

Before every save OreStudio prompts for a *change reason* — a structured code identifying the business justification. Change reasons are organised into categories accessible from *Reference Data* → *Audit Trail*.



Code	Description	Version	Modified By	Recorded At
trade	Trade lifecycle reasons aligned with FINRA and MiFID II standards	1	ores_local_d...	40 minutes...
system	System-generated reasons for automatic operations (not user-selectable)	1	ores_local_d...	40 minutes...
common	Universal data quality reasons aligned with BCBS 239 and FRTB standards	1	ores_local_d...	40 minutes...

Figure 4.3: The Change Reason Categories window showing the three built-in category groups and their regulatory alignment.

Screenshot pending: the Change Reasons list window (*Reference Data* → *Audit Trail* → *Change Reasons*).

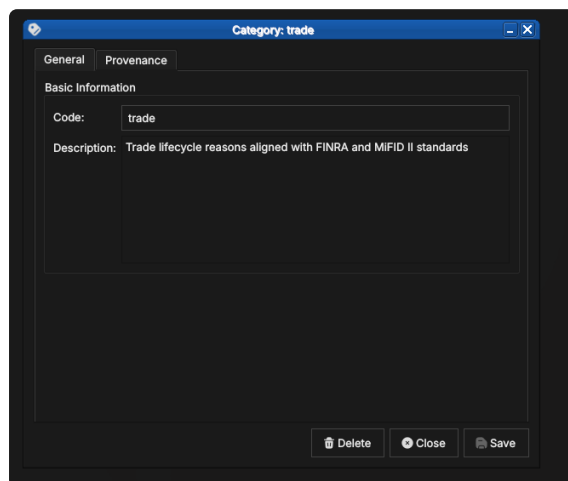


Figure 4.4: The detail dialog for the trade category, showing its regulatory description.

Change reasons serve two purposes: they act as a forcing function against silent undocumented changes, and they make the audit trail machine-readable and regulatorily aligned. The category structure maps directly to BCBS 239, FRTB, MiFID II, and FINRA reporting obligations.

1. The System Category

System reasons are assigned automatically by the application and services. They are not shown in the change reason prompt and cannot be selected manually.

(a) `system.initial_load`

Applies to	Create
Commentary	Optional

Initial system provisioning or database migration. Used once during deployment.

(b) `system.new_record`

Applies to	Create
Commentary	Optional

Normal operational record creation by the application or a service.

(c) `system.external_data_import`

Applies to	Amend
Commentary	Mandatory

Import from an external data source (ISO feed, GLEIF, vendor file). Commentary is mandatory — the data lineage source must be recorded.

(d) `system.import`

Applies to	Create, Amend
Commentary	Optional

Data loaded via the CLI import command.

(e) `system.test`

Applies to	Create, Amend, Delete
Commentary	Optional

Test data created by automated test suites.

(f) `system.tenant_terminated`

Applies to	Amend
Commentary	Optional

Applied when a tenant is marked as terminated.

(g) `system.admin_reset`

Applies to	Amend
Commentary	Optional

Data reset by a system administrator during re-provisioning.

2. The Common Category

Common reasons are the universal data quality reasons, aligned with BCBS 239 and FRTB standards. These are the reasons most frequently used by operators during day-to-day data maintenance.

(a) `common.non_material_update`

Applies to	Amend
Commentary	Optional

A cosmetic or administrative change with no economic impact — a "touch" to refresh a timestamp or correct capitalisation.

(b) `common.rectification`

Applies to	Amend, Delete
Commentary	Optional

Correction of a user or booking error. The most commonly used reason for fixing a data-entry mistake.

(c) `common.duplicate`

Applies to	Delete
Commentary	Optional

Removal of a duplicate record where an equivalent entry already exists.

(d) `common.stale_data`

Applies to	Amend, Delete
Commentary	Optional

Data not updated within the required liquidity horizon, requiring a refresh or removal.

(e) `common.outlier_correction`

Applies to	Amend, Delete
Commentary	Mandatory

Manual override following a plausibility check failure — a value outside acceptable bounds overridden by an operator. Commentary must identify the plausibility rule that triggered the review.

(f) `common.feed_failure`

Applies to	Amend, Delete
Commentary	Mandatory

Correction caused by an upstream vendor or API data issue. Commentary must identify the failed feed.

(g) `common.mapping_error`

Applies to	Amend, Delete
Commentary	Mandatory

Incorrect identifier translation — for example a wrong ISIN-to-FIGI mapping. Commentary must describe the mapping error.

(h) `common.judgmental_override`

Applies to	Amend, Delete
Commentary	Mandatory

Expert judgment applied when market prices or reference values are unavailable and an operator must supply a value manually.

(i) `common.regulatory`

Applies to	Amend, Delete
Commentary	Mandatory

Mandatory compliance adjustment required by a regulatory obligation or instruction.

(j) `common.other`

Applies to	Amend, Delete
Commentary	Mandatory

Exceptional changes that do not fit any other category. The commentary must fully explain the reason — this code exists as a last resort and should be used sparingly.

3. The Trade Category

Trade reasons are aligned with FINRA and MiFID II trade lifecycle reporting requirements.

(a) `trade.fat_finger`

Applies to	Amend, Delete
Commentary	Optional

Erroneous execution — a trade entered with the wrong quantity, price, or instrument due to a keying error.

(b) `trade.system_malfunction`

Applies to	Amend, Delete
Commentary	Mandatory

Change caused by a technical glitch or algorithmic error in an execution system. Commentary must identify the system and the nature of the malfunction.

(c) `trade.corporate_action`

Applies to	Amend, Delete
Commentary	Optional

Adjustment following a corporate action such as a stock split, dividend reinvestment, or merger.

(d) `trade.allocation_swap`

Applies to	Amend, Delete
Commentary	Optional

Reallocation between a house account and a client sub-account, or between sub-accounts.

(e) `trade.re_booking`

Applies to	Amend, Delete
Commentary	Mandatory

Correction of a wrong legal entity booking — a trade entered under the wrong counterparty or book. Commentary must identify the correct entity.

(f) `trade.other`

Applies to	Amend, Delete
Commentary	Mandatory

Exceptional trade lifecycle changes that do not fit any other code.

4. Extending Change Reasons

Change reason categories and individual codes are stored as versioned reference data and can be added through the same Qt UI and CLI tools used for other entities. In principle, a tenant administrator can create additional categories and codes for firm-specific workflows.

In practice this should be done cautiously:

- The standard codes are aligned with regulatory frameworks (BCBS 239, FRTB, MiFID II, FINRA). Non-standard codes risk creating audit trail entries that do not map cleanly to regulatory reports.
- `common.other` and `trade.other` with mandatory commentary cover most exceptional cases without adding custom codes.
- Custom codes cannot be distinguished from standard codes by the system, so the tenant must maintain its own record of which codes are standard and which are custom.

- Removing or renaming a code after it has been used in the audit trail leaves historical records referencing a code whose meaning is no longer documented.

See also

- [ISO 8000](#) — international standard for data quality.
- [DAMA-DMBOK](#) — data management body of knowledge, including the DQ-6 framework.
- [Temporal database](#) — background on transaction time, valid time, and the bitemporal model.
- *Time and Timestamps: Architecture and Conventions* — internal architecture document ([doc/knowledge/architecture/time-architecture.org](#)).
- [BCBS 239](#) — Basel Committee principles for effective risk data aggregation and reporting.
- [GLEIF](#) — the Global Legal Entity Identifier Foundation; publishes the LEI registry and the LEI-to-BIC mapping dataset.

Currencies

This chapter covers currency management end to end: the ISO 4217 standard and OreStudio's extensions, every screen in the Qt UI, and the three classification tables currencies depend on — rounding types, market tiers, and monetary natures.

Overview

CURRENCIES ARE the foundational reference data entity. Every trade, cashflow, position, and P&L figure in the system is denominated in a currency; getting the currency record right is a prerequisite for everything else.

ISO 4217 and its limitations

The international standard for currency codes is ISO 4217. It defines a three-letter alphabetic code (USD, EUR, GBP), a three-digit numeric code (840, 978, 826), a currency name, and the number of decimal places — for example, USD has 2 minor units, meaning amounts are expressed to the nearest cent.

ISO 4217 is widely adopted and OreStudio uses it as its primary identifier. However, the standard has gaps that matter in practice:

- It does not classify currencies by **market tier** — the distinction between major G10 currencies and emerging-market currencies is economically important but absent from the standard.
- It does not cover **digital assets**. Cryptocurrencies such as Bitcoin or Ethereum are increasingly relevant in financial systems but have no ISO 4217 code.
- It says nothing about **display formatting** — whether a symbol precedes or follows the amount, what the thousands and decimal separators are, or how the fractional unit is labelled.

- It does not specify **rounding conventions** — whether a calculation result should be rounded up, down, or to the nearest unit.

OreStudio extends the ISO 4217 model with additional fields to cover these gaps: monetary nature, market tier, currency symbol, fraction symbol, a printf-style format string, and configurable rounding rules. The standard fields remain primary; the extensions are supplementary.

The Currencies window

Open the Currencies window from the *Reference Data* menu. It lists all currencies defined in the tenant, with one row per currency.

Code	Currency Name	Numeric Code	Symbol	Version	Modified By	Recorded At
AED	UAE Dirham	784	ﷵ	1	ore_local_refdata_service	2 minutes ago
AFN	Afghan Afghani	971	ؑ	1	ore_local_refdata_service	2 minutes ago
ALL	Albanian Lek	008	L	1	ore_local_refdata_service	2 minutes ago
AMD	Armenian Dram	051	֏	1	ore_local_refdata_service	2 minutes ago
ANG	Netherlands Antillean Guilder	532	ƒ	1	ore_local_refdata_service	2 minutes ago
AUD	Australian Dollar	036	\$	1	ore_local_refdata_service	2 minutes ago
AUG	Aruban Florin	533	ƒ	1	ore_local_refdata_service	2 minutes ago
AZN	Azerbaijani Manat	944	ⴖ	1	ore_local_refdata_service	2 minutes ago
BAM	Bosnia and Herzegovina Convertible Mark	977	KM	1	ore_local_refdata_service	2 minutes ago
BBD	Barbadian Dollar	052	\$	1	ore_local_refdata_service	2 minutes ago
BDT	Bangladesh Taka	050	৳	1	ore_local_refdata_service	2 minutes ago
BGN	Bulgarian Lev	975	лв	1	ore_local_refdata_service	2 minutes ago
BMD	Bermudian Dollar	060	\$	1	ore_local_refdata_service	2 minutes ago
BND	Brunei Dollar	096	B\$	1	ore_local_refdata_service	2 minutes ago
BOB	Bolivian Boliviano	068	Bs	1	ore_local_refdata_service	2 minutes ago
BSD	Bahamian Dollar	044	\$	1	ore_local_refdata_service	2 minutes ago
BTN	Bhutanese Ngultrum	064	Nu.	1	ore_local_refdata_service	2 minutes ago
BYN	Belarusian Ruble	933	Br	1	ore_local_refdata_service	2 minutes ago
BZD	Belize Dollar	084	BZ\$	1	ore_local_refdata_service	2 minutes ago
CHF	Swiss Franc	756	CHF	1	ore_local_refdata_service	2 minutes ago
CNH	Offshore Chinese Yuan (Hong Kong)		¥	1	ore_local_refdata_service	2 minutes ago

Figure 5.1: The Currencies window, showing the full list of currencies in the tenant. Each row shows the ISO code, name, numeric code, symbol, version number, the identity of the last modifier, and when the record was last recorded. The status bar shows the current page and total record count.

The toolbar buttons reload the list and adjust the display. The list is paginated; use the page controls at the bottom right to navigate. Double-clicking a row opens the **Currency Details** dialog.

Currency Details

The Currency Details dialog has four tabs: **General**, **Formatting**, **Rounding**, and **Provenance**. The three action buttons at the bottom — **Delete**, **Close**, and **Save** — apply to the currency as a whole.

General

The General tab carries the core identity of the currency:

- **ISO Code** — the three-letter ISO 4217 alphabetic code. This is the primary key and cannot be changed after creation.
- **Name** — the full English name of the currency (e.g. *Angolan Kwanza*).
- **Numeric Code** — the ISO 4217 numeric code (e.g. 973 for AOA).
- **Asset Class** — the monetary nature of the currency: `fiat` for a government-issued currency, `crypto.major` for a major digital asset. See [Monetary Natures](#) below for the full list.

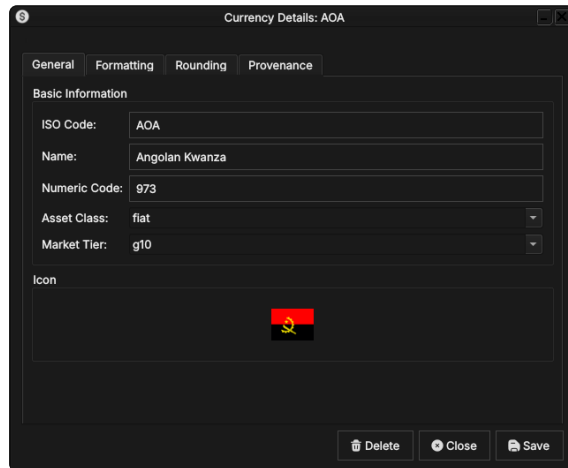


Figure 5.2: The General tab for the Angolan Kwanza (AOA). It shows the ISO code, full name, numeric code, monetary nature, market tier, and the country flag icon associated with the currency.

- **Market Tier** — the currency’s liquidity classification: g10 for the major currencies, emerging for others. See [Market Tiers](#) below.
- **Icon** — the country flag for the currency’s issuing nation, populated automatically from the system’s image dataset on import.

Formatting

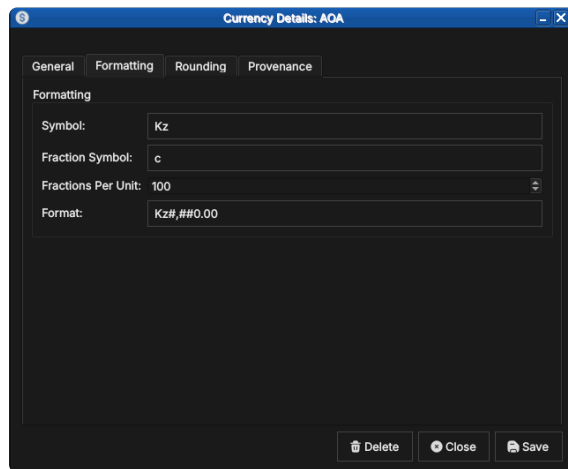


Figure 5.3: The Formatting tab for AOA, showing the currency symbol (Kz), the fractional unit symbol (c for centavo), fractions per unit (100), and the display format string.

The Formatting tab controls how currency amounts are displayed:

- **Symbol** — the currency symbol, e.g. Kz for the Kwanza or \$ for the US Dollar.
- **Fraction Symbol** — the symbol for the fractional unit, e.g. c for centavo.
- **Fractions Per Unit** — the number of fractional units in one whole unit. Most fiat currencies use 100; the Kuwaiti Dinar uses 1000; currencies with no fractional unit (e.g. Japanese Yen) use 0; cryptocurrencies typically use 100000000.
- **Format** — a printf-style format string controlling amount rendering, e.g. Kz#,##0.00 produces Kz1,234.56.

Rounding

The Rounding tab controls how calculated amounts are rounded before storage and display:

- **Rounding Type** — the rounding algorithm applied to amounts in this currency. See [Rounding Types](#) below for the full list of options.
- **Rounding Precision** — the number of decimal places to round to. Typically 2 for most fiat currencies and 8 for cryptocurrencies.

Provenance

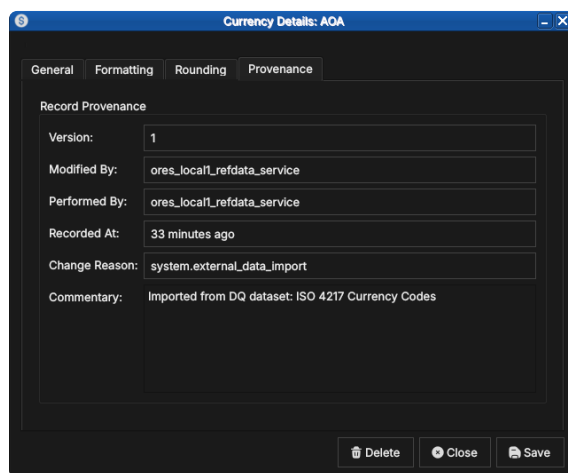


Figure 5.4: The Provenance tab showing record metadata: version number, the service that last modified the record, who performed the operation, when it was recorded, the change reason code, and a free-text commentary.

The Provenance tab is read-only and shows the audit metadata for the current version. See the [Reference Data — Provenance](#) section for a full description of the provenance fields common to all reference data entities.

Editing a currency

To edit a currency, open it in the Currency Details dialog, make your changes, and click **Save**. Before the record is written, OreStudio prompts for a change reason.

Select a **Reason** from the drop-down and add optional **Commentary**. Click **Save** to confirm. Every save creates a new version; the previous version is never overwritten.

Currency history

To view the full change history, open the details dialog and click the history icon in the title bar, or right-click the row and choose **History**.

Select a version to populate the detail panel. The **Changes** tab shows which fields changed between the selected version and its predecessor. The **Revert** button reinstates any historical version as a new version, preserving the full audit chain.

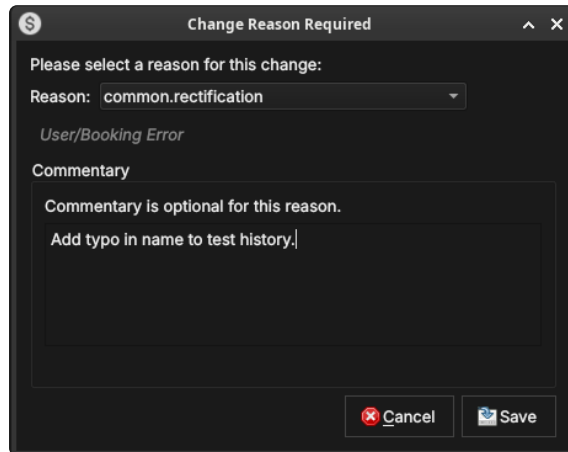


Figure 5.5: The Change Reason Required dialog, which appears before every save. A reason code must be selected; commentary is optional for some reason codes and required for others.

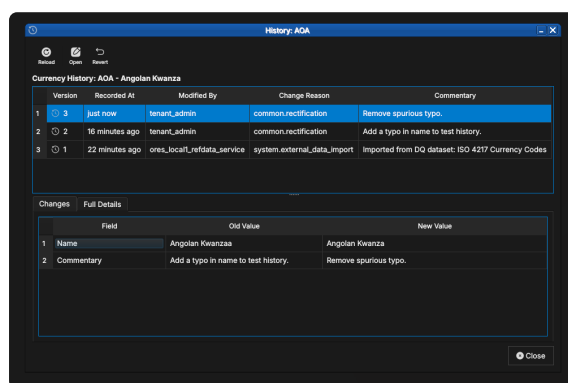


Figure 5.6: The History dialog for the Angolan Kwanza, showing three versions: the original ISO 4217 import (version 1), a test correction (version 2), and a final rectification (version 3, currently selected). The Changes tab shows that only the Name and Commentary fields changed between version 2 and version 3.

Auxiliary Data

The three tables described in this section act as classification vocabularies for currencies. In a standard deployment their values are fixed at provisioning time and rarely change; OreStudio still versions and audits them the same way as any other reference data. They are managed by the system tenant and are accessible from the *Reference Data* → *Currencies* sub-menu.

Rounding Types

Rounding types specify the algorithm used to round currency amounts in financial calculations. The values match the `roundingType` enumeration in ORE's XML schema, ensuring consistency with any ORE configuration files loaded into the system.

Open the Rounding Types window from *Reference Data* → *Currency Codes* or from the toolbar within the Currency Detail dialog.

Double-clicking a row opens the detail dialog, which shows the code, name, and a description that includes numeric examples at two decimal places.

There are five standard values:

1. Up

Rounds away from zero regardless of the fractional part. The amount is always moved to the next representable value in the

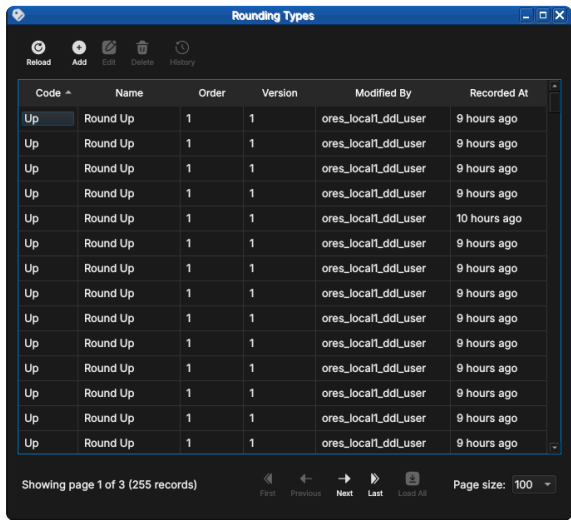


Figure 5.7: The Rounding Types list window showing the five standard rounding methods with their display order, version, and provenance.

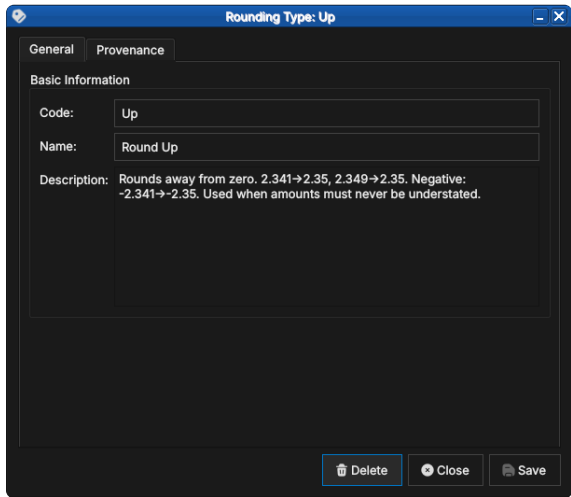


Figure 5.8: The detail dialog for the Up rounding type, showing the code, human-readable name, and a description with concrete examples.

direction away from zero.

Examples at 2 decimal places: $2.341 \rightarrow 2.35$, $2.349 \rightarrow 2.35$. For negative amounts: $-2.341 \rightarrow -2.35$.

Use Up when amounts must never be understated — for example, when calculating a fee that must cover the full cost.

2. Down

Truncates toward zero. The fractional part is discarded.

Examples: $2.349 \rightarrow 2.34$, $-2.341 \rightarrow -2.34$.

Use Down in conservative contexts where overstating an amount would be the more serious error — for example, when reporting a liability that should not be exaggerated.

3. Closest

Rounds to the nearest representable value, with halves rounded away from zero. This is the most common rounding mode for financial amounts and is the default for most fiat currencies.

Examples: $2.344 \rightarrow 2.34$, $2.345 \rightarrow 2.35$, $-2.345 \rightarrow -2.35$.

4. Floor

Always rounds toward negative infinity — i.e., to the next lower value regardless of sign. Unlike Down, which rounds toward zero, Floor produces different results for negative amounts.

Examples: $2.349 \rightarrow 2.34$, $-2.341 \rightarrow -2.35$.

Use Floor when rounding must never produce a value higher than the unrounded input, even for negative amounts.

5. Ceiling

Always rounds toward positive infinity — i.e., to the next higher value regardless of sign.

Examples: $2.341 \rightarrow 2.35$, $-2.349 \rightarrow -2.34$.

Use Ceiling when rounding must never produce a value lower than the unrounded input.

Market Tiers

Market tiers classify currencies by liquidity profile and market accessibility. The classification feeds into margin calculations, settlement conventions, and risk limits in downstream analytics.

Open the Currency Market Tiers window from *Reference Data* → *Classifications*.

There are five standard tiers:

1. G10 / Majors

The ten most liquid and widely traded global currencies: USD, EUR, GBP, JPY, CHF, AUD, NZD, CAD, SEK, and NOK. G10 currencies have deep interbank markets, tight bid/offer spreads, and continuous 24-hour liquidity. They are the primary currencies for

Code	Name	Order	Version	Modified By	Recorded At
historical	Historical	5	1	ores_local_iam_service	11 minutes ago
g10	G10 / Majors	1	1	ores_local_iam_service	11 minutes ago
frontier	Frontier	4	1	ores_local_iam_service	11 minutes ago
exotic	Exotic	3	1	ores_local_iam_service	11 minutes ago
emerging	Emerging	2	1	ores_local_iam_service	11 minutes ago

Workspace: **Live**

Showing all records (5) Page size: 100

Figure 5.9: The Currency Market Tiers list showing all five standard tiers — historical, g10, frontier, exotic, and emerging — with their display order and provenance.

Currency Market Tier: historical

General | Provenance

Basic Information

Code: historical

Name: Historical

Description: Currencies no longer in active use

Delete Close Save

Figure 5.10: The detail dialog for the historical tier, showing the code, name, and description.

derivatives trading and the dominant funding and collateral currencies in global finance.

2. Emerging

Currencies from developing economies with growing but still maturing financial markets. They typically have moderate liquidity, wider spreads than G10, and may be subject to capital controls or central bank intervention. Examples include BRL (Brazilian Real), MXN (Mexican Peso), ZAR (South African Rand), INR (Indian Rupee), and CNY (Chinese Renminbi).

3. Exotic

Thinly traded currencies with wide bid/offer spreads and limited liquidity outside their domestic market. Forward markets may be shallow or absent. Examples include currencies of smaller African, Central Asian, or Pacific island nations.

4. Frontier

Currencies from frontier markets with limited or restricted convertibility. These may be pegged to a major currency, subject to official exchange rates, or carry transfer restrictions that make them impractical for cross-border settlement. Examples include currencies from markets classified as frontier by MSCI or FTSE Russell.

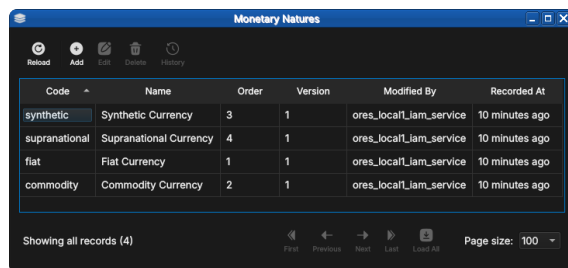
5. Historical

Currencies no longer in active use. This includes legacy eurozone currencies superseded by the Euro (DEM, FRF, ITL, ESP, and others), as well as currencies from defunct states or monetary unions. Historical currencies are retained in the system for back-office reconciliation and historical analytics on older trade populations.

Monetary Natures

Monetary nature classifies currencies by the underlying economic mechanism that gives them value. The classification matters for collateral eligibility, regulatory capital treatment, and the applicability of pricing models.

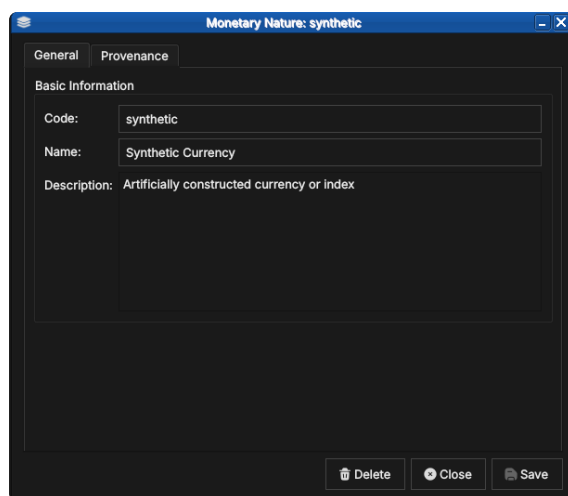
Open the Monetary Natures window from *Reference Data* → *Classifications*.



Code	Name	Order	Version	Modified By	Recorded At
synthetic	Synthetic Currency	3	1	ores_local_iam_service	10 minutes ago
supranational	Supranational Currency	4	1	ores_local_iam_service	10 minutes ago
fiat	Fiat Currency	1	1	ores_local_iam_service	10 minutes ago
commodity	Commodity Currency	2	1	ores_local_iam_service	10 minutes ago

Showing all records (4)

Figure 5.11: The Monetary Natures list showing all four standard values — synthetic, supranational, fiat, and commodity — with their display order and provenance.



Monetary Nature: synthetic

General Provenance

Basic Information

Code: synthetic

Name: Synthetic Currency

Description: Artificially constructed currency or index

Delete Close Save

Figure 5.12: The detail dialog for the synthetic monetary nature, showing the code, name, and description.

There are four standard values:

1. Fiat

Government-issued currency that derives its value from legal tender status rather than from a physical commodity. The issuing central bank or treasury manages supply through monetary policy. The overwhelming majority of currencies in the ISO 4217 standard are fiat — USD, EUR, GBP, JPY, and so on.

2. Commodity

Currency backed by or representing a claim on a physical commodity. In practice this covers the ISO 4217 commodity codes:

XAU (gold), XAG (silver), XPT (platinum), and XPD (palladium). Pricing is driven by spot commodity markets rather than interest rate differentials.

3. Synthetic

Artificially constructed currency or index that does not correspond to any single government or commodity. Examples include basket currencies and internal transfer pricing units used within large financial institutions.

4. Supranational

Currency issued by a multi-national authority rather than a single sovereign. The primary example is XDR (Special Drawing Rights), issued by the International Monetary Fund as an international reserve asset and valued as a basket of USD, EUR, CNY, JPY, and GBP — reviewed and revised by the IMF every five years.

Shell commands

The ORE Studio interactive shell provides quick currency access without opening the Qt UI:

```
# List all currencies (paginated)
currencies get

# Add a new currency: <iso_code> <name> <numeric_code> <symbol> <fractions_per_unit>
currencies add XTS "Test Currency" 963 T$ 100

# Delete a currency by ISO code
currencies delete XTS

# Show history for a currency
currencies history USD
```

CLI commands

The `ores.cli` command-line tool provides a richer interface for automation. All currency commands live under `ores.cli reldata currencies`:

```
# List all currencies as a table
ores.cli reldata currencies list --format table

# List a specific currency as JSON
ores.cli reldata currencies list --format json --key EUR

# Add a new currency
ores.cli reldata currencies add \
  --iso-code XTS --name "Test Currency" \
  --numeric-code 963 --modified-by operator \
  --currency-type fiat
```

```
# Delete a currency by ISO code  
ores.cli refdata currencies delete --iso-code XTS
```

```
# Export all currencies to a file  
ores.cli refdata currencies export --output currencies.json
```

See also

- [ISO 4217 Currency Codes](#) — the international standard for currency identification.
- [SIX Financial Information](#) — the ISO 4217 maintenance agency.
- [MSCI Market Classification](#) — framework for developed, emerging, and frontier market designations.